Risk Aversion and Barriers to Firm Growth: Experimental Evidence from Small Retailers*

Grady Killeen

The University of Chicago

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Abstract

Does risk aversion prevent firms in developing economies from making profitable but uncertain investments? Using a model and two field experiments with over 1,200 Kenyan firms, I study retailers' decisions to stock a new consumer product. First, an insurance contract that is strictly dominated under risk neutrality increases adoption from 20% to 30%. Second, a temporary supplier returns policy increases post-intervention stocking by 70%, through reduced belief uncertainty rather than higher expected profits. Third, increasing the continuation value of learning expands adoption by 88%. Results indicate that risk aversion constrains firm experimentation, limiting the retail diffusion of new products.

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1 Introduction

Firms in low and middle-income countries (LMICs) often grow slowly and have low productivity (Bloom et al., 2010). Much of the existing literature on firms, including small firms in poor countries, assumes risk neutrality, meaning only expected returns should drive decisions. Yet most production in developing economies occurs among small enterprises where owners bear a significant share of profits and losses. In the absence of complete insurance markets, owners' consumption may be sensitive to business performance, causing their risk preferences to affect production decisions. This may deter enterprises from engaging in risk taking required to grow.

This paper tests whether risk aversion causes small firms to forgo profitable but risky investments. The returns to new business practices, such as employing a new technology or stocking a new product, are inherently uncertain. If firms are risk averse, they may fail to engage in experimentation to identify profitable opportunities, limiting innovation and growth.

I explore this question in the context of retail firms' decision to stock a new consumer product. Offering new products can increase retailers' profits. But shops are often uncertain about whether consumers will demand new goods, creating a risk that inventory investments will lead to losses. This setting also has direct implications for upstream producers. A manufacturer who cannot secure retail distribution faces a smaller effective market, reducing the returns to introducing new goods.

My empirical setting is the Kenyan market for new motorcycle helmets. In 2020, a corporation built a USD $3 million Kenyan factory and began producing one of the first effective helmets affordable to consumers, selling at a wholesale price of USD PPP 15 (high-quality imports cost $100 or more). But their retail diffusion was slow. Two years after their introduction, over 50% of shops near the factory believed that selling helmets would yield positive expected profits, yet only 7% stocked them. Although many firms held optimistic expectations, their beliefs were often diffuse, and retailers did not typically stock the good if they were unsure about its profitability.

I begin by developing a model that draws stark contrasts between the experimentation and learning behavior of risk neutral versus risk averse firms when faced with an uncertain investment. In the model, an entrepreneur decides whether to stock a new good with unknown demand. This embeds a bandit learning problem into a standard model of small firm behavior, generating predictions identifying risk aversion and demand uncertainty. The core insight is that when firms are more uncertain

about a product's profitability, risk neutral firms invest more because upside is high. This ensures that risk neutral retailers quickly discover profitable new products. In contrast, risk aversion creates disutility from uncertainty, which can lead firms to invest less in products the more uncertain they are about their profitability, undermining new product adoption.

Guided by the model, I design two field experiments to test if risk aversion prevents enterprises from adopting new products. The first experiment, the Insurance Experiment, constructs a precise test for whether risk aversion affects firms' stocking decisions. The second experiment, the Learning Experiment, isolates the importance of risky experimentation to learn in an uncertain environment. This experiment shows that temporarily lowering the risk of experimenting leads to sustained increases in helmet stocking. I leverage a treatment arm and model predictions to argue that persistence is driven by risk averse firms overcoming uncertainty and cannot readily be explained by confounding explanations such as incorrect beliefs or learning by doing.

The Insurance Experiment (N=350 firms) tests if risk aversion affects firms' stocking decisions. I offered shops the opportunity to stock helmets with or without an insurance contract designed to be strictly dominated under risk neutrality but valuable under risk aversion. I test if enterprises are risk neutral by examining the effect of this contract on stocking.

Testing firms' risk preferences is complicated by possible capital constraints, since insurance contracts typically charge upfront premiums, affecting liquidity. To isolate risk aversion, my design offered control shops an unconditional future payment. Randomly selected *insurance* shops chose between this payment and a contract paying more if they stocked helmets and failed to sell out, but nothing otherwise. Offers were calibrated so that, given firms' beliefs, the contract's expected value fell below the fixed payment, creating a mean-preserving (technically decreasing) contraction of helmet profits. Therefore, *insurance* should increase stocking only if firms are risk averse.

The rate of *insurance* shops that immediately stocked helmets was 7 percentage points (160%, $p = .01$) higher. Firms were given up to two weeks to order to search for customers. The *insurance* effect was 10pp (50%, $p = .02$) after this period. This implies a rejection of risk neutrality. The results are not driven only by new or unproductive firms: effects remained large and significant among firms in business for at least 4 years, with employees, and with above median profits.

The design is robust to two additional confounds. First, surveyors followed a script to inform all shops about *insurance* and to explain that the shops receiving the offer would be selected at

random. Then the shop's treatment assignment was revealed. This prevents *insurance* from sending a signal about demand. Second, the contract included conditions ensuring that stocking helmets and intentionally failing to sell out to obtain the insurance payout was never optimal.

The Learning Experiment ($N = 929$ firms) tests if risk aversion prevents firms from engaging in experimentation needed to discover new products. The design builds on the model's prediction that firms face a risk-reward trade-off: by stocking a new product, they risk short-term losses but may gain long-term profits by restocking the product if they learn that it is profitable. The sample consists of firms that could have acquired helmets from the manufacturer for over two years but did not stock them. Two treatments test whether risk aversion prevented firms from learning about helmets' profitability in this setting where their background diffusion was slow.

The first *returns* treatment was designed to test if temporarily lowering stocking risk caused firms to learn about demand and continue stocking helmets. All firms received access to helmet stock at market prices. The intervention had two phases. In phase one, *returns* shops could return unsold stock for a refund whereas control shops could not. In phase two, all firms were given the option to buy stock without buybacks, so that offers were identical for *returns* and control shops. The key test is whether the one-time reduction in risk led to increases in phase two stock purchases.

Consistent with the Insurance Experiment, the temporary reduction in inventory risk led to a large increase in helmet uptake in phase one: *returns* increased phase one uptake by 9 percentage points relative to a 6.8% control rate (128%, $p < .01$). Reflecting firms' baseline expectations that the product would be profitable, few firms exercised the return option and 90% of shops directly reported that they were profitable, with magnitudes amounting to 10% of total firm profits.

Initial uptake differences translated into persistently higher rates of helmet stocking in phase two, when all firms faced the same stocking conditions. *Returns* shops were 7 percentage points (70%, $p = .02$) more likely to purchase helmet stock (from any supplier). This was driven by high restocking rates consistent with reported profitability: $2/3$ of phase one adopters restocked and 80% planned to stay in the market. The increase in phase two stocking is consistent with *returns* inducing risk averse firms to try selling helmets, who then resolved demand uncertainty by learning.

A confounding explanation is that shops held incorrect beliefs that helmets were unprofitable, and the intervention positively updated their expectations. The model predicts divergent patterns in firms' beliefs under these two mechanisms, which I test using novel firm survey data.

If *returns* mitigate risk aversion, (1) belief variance should negatively predict uptake among untreated firms (stocking is perceived as risky), (2) *returns* should induce stocking among firms uncertain about demand, and (3) beliefs should become more precise with experience. Under the alternative of changing mean beliefs, (1) uncertainty should positively predict uptake among untreated firms (information is more valuable), (2) the treatment should attract firms with pessimistic priors, and (3) beliefs should become more optimistic with experience. The empirical evidence supports the importance of risk aversion: all three predictions associated with this mechanism hold, versus none under the alternative of positively updating expectations.

These results highlight the importance of firm learning under uncertainty. When experimentation is risky but can generate long-run gains, a firm's willingness to experiment depends not only on risk but also on the perceived value of future opportunities. This implies an additional prediction of the model: firms should be more willing to undertake experimentation in the present when they anticipate larger payoff potential in the future, even if current-period payoffs are unchanged.

The Learning Experiment tests this hypothesis using a second *supplier commitment* treatment. Firms expressed concerns that the manufacturer may stop supplying small shops in the future, reducing future upside from discovering if helmets were profitable. I therefore offered, during phase one, to help *supplier commitment* shops restock directly from the supplier after the study. As in the Insurance Experiment, control shops were aware of this arm but informed that they would need to identify a supplier themselves. *Supplier commitment* firms were 6 percentage points (88%, $p = .03$) more likely to stock helmets during phase one. This shows that firms are forward-looking, suggesting that firms internalize the continuation value of learning into their stocking decisions.

The model further predicts that the *supplier commitment* should not increase stocking among firms offered *returns* because *returns* should induce all firms constrained by demand uncertainty to stock. Point estimates match this prediction: the *supplier commitment* has little additional effect conditional on receiving *returns*.[1] This weighs against confounding forces, such as learning by doing, that would typically cause the *supplier commitment* to affect uptake even with *returns*.

Another implication of demand uncertainty as the key source of risk constraining helmet adoption is that information from other sources should affect stocking. Evidence of information

[1]This difference is not statistically significant overall, but is among firms that did not know of a seller at baseline, the one pre-registered heterogeneity dimension.

spillovers supports this view. First, *returns* had no effect among firms near an existing seller, where uptake was common in all arms. Second, giving a random group of firms peers' sales data increased stocking by 2 percentage points ($p = .02$). Third, I induced sellers to enter randomly selected markets and later offered a new *spillover* sample helmet stock in these and control markets. Firms in markets with an induced seller were nearly twice as likely to stock ($p < .01$).

These results indicate that risk aversion may constrain the retail market for new products, which could harm upstream producer incentives. *Returns* shops stocked twice as many helmets (3.4 vs 1.5, $p = .03$) and were twice as likely to restock and report intent to keep selling helmets at endline (13.4% vs 6.8%, $p = .01$). These results need not imply market expansion if they displaced other shops' sales. Two results suggest that such displacement is unlikely. First, *returns* increased the rate at which a shop reported any local seller at endline, including itself, by 7.8pp ($p = .04$). Second, the spillover results indicate that a firm stocking helmets *crowds in* other sellers.

This study contributes to a literature studying barriers to firm growth in developing countries. Studies have examined how inputs such as capital (de Mel et al., 2008), labor (de Mel et al., 2019), and management (Bloom et al., 2013) affect productivity. This paper shows that risk aversion can deter firms from engaging in risk taking needed to grow, preventing retailers from adopting a new product. Prior work relates measures of owners' risk preferences to firm outcomes (de Mel et al., 2008; Kremer et al., 2013) and examines how loan terms easing repayment or sharing downside risk affect entrepreneurs' investments (Field et al., 2013; Fischer, 2013; Meki, 2025). However, risk preferences may correlate with other owner characteristics, and changing loan terms affects liquidity needs and investment returns, so neither approach cleanly isolates risk aversion. The resulting evidence is also mixed, leaving open whether risk aversion is a first-order determinant of small firm behavior. I lower investment risk without increasing expected returns, providing unambiguous evidence that risk aversion can consequentially affect firms' investment decisions.

This paper also builds on research studying risk in agriculture. Studies find that receipt of insurance (Karlan et al., 2014; Cole et al., 2017) or risk-lowering technology (Emerick et al., 2016) affects farmers' production choices. Non-farm firms are distinct because risk neutrality is often assumed to approximate their behavior, in part because they operate in competitive markets that may discipline first-order deviations from profit maximization (Kremer et al., 2019).[2] Their risks

[2]Kremer et al. (2019) describe competition as a longstanding justification for treating firms as profit maximizers,

are also often idiosyncratic and difficult to contract on, so products like index insurance are not available as a tool to study their risk preferences. This study overcomes this challenge by contracting directly on firms' own sales and calibrating the contract to their subjective beliefs.

These findings imply that the common practice of modeling small firms as risk neutral may not be appropriate. Empirical tests across many topics in development economics, including capital constraints, collusion, and technology adoption, rely on profit maximization as a simplifying assumption that abstracts away from possible risk aversion (Banerjee and Duflo, 2014; Bergquist and Dinerstein, 2020; Bassi et al., 2022). The interpretation of these tests may change if risk aversion creates first-order deviations from profit maximization. For instance, the marginal product of capital may vary across firms due to heterogeneous risk preferences, not only borrowing access.

This paper also contributes to a literature on firm upgrading and technology adoption in LMICs. Verhoogen (2023) summarizes evidence from manufacturers showing that factors including organizational structure (Atkin et al., 2017a), export exposure (Atkin et al., 2017b), and access to intermediate inputs (Goldberg et al., 2010) affect upgrading. Less is known about retailers, who play a direct role in product diffusion. I show that risk aversion prevents retailers from stocking new products, a simple yet common form of innovation (de Mel et al., 2009). This relates to work showing that high search costs can prevent importers from adopting new products (Startz, 2025).

A further contribution is to research on firm learning about demand. My model builds on theoretical work on demand uncertainty (Rothschild, 1974); I show that risk aversion can prevent firm experimentation. This helps explain why firms can hold misperceptions about demand (Bai et al., 2025), and complements evidence that customers are an important channel for retailer learning (Argente et al., 2026). The paper also relates to research on agricultural technology adoption, which has emphasized learning from others (Besley and Case, 1993; Foster and Rosenzweig, 1995; Conley and Udry, 2010), and to research showing that temporary subsidies can persistently increase technology adoption (Carter et al., 2021; Dupas, 2014). I emphasize uncertainty as a dynamic incentive to experiment, which highlights why risk aversion can prevent learning, similar to an argument that Bryan et al. (2014) make in the case of migration. Finally, I separately vary experimentation costs and the future value of learning to test the risk-reward trade-off of bandit models.

though they argue it may be less likely to hold in LMICs. Macroeconomic studies more often allow for risk averse entrepreneurs (e.g. Buera et al., 2011), though profit maximization is still a common simplifying assumption.

2 A model of small firm learning about demand

In this section, I model an entrepreneur faced with the decision of whether or not to stock a new product. I use the model to construct tests of risk aversion and learning about demand that guide the design and interpretation of the experiments.

2.1 Model setup

I consider an infinitely repeated, single agent dynamic optimization problem in discrete time. The model is single agent because the hypotheses studied are not strategic. The problem is dynamic, as learning enables agents to refine their future decisions.

The entrepreneur's problem: An entrepreneur chooses how much of a safe good $j = s$ and a new good $j = n$ to stock in each period t. Inventory is sold in $t + 1$. The demand curve of the safe good $p_s(q_{st}, \nu_{st}) = \alpha_s(q_{st}) + \nu_{st}$ is known, but realizations are subject to iid fluctuations $\nu_{st} \sim \mathcal{N}(0, \Sigma_s)$.

Demand for the new product, $p_n(q_{nt}, \nu_{nt} + \theta) = \alpha_n(q_{nt}) + \nu_{nt} + \theta$, is indexed by a parameter $\theta \in \mathbb{R}$. The true value, θ_0, is unknown to the agent. $\nu_{nt} \sim \mathcal{N}(0, \Sigma_n)$ again captures iid fluctuations in demand. These influence the rate at which agents learn and allow for a distinction between two sources of uncertainty: diffuse prior beliefs, which can be resolved through learning, and stochastic demand fluctuations, which introduce variance in profits even when the demand curve is known. The demand curves are continuously differentiable and downward sloping $\left(\frac{\partial p_j}{\partial q_{jt}} \leq 0 \; \forall q_{jt}, \nu_{jt}\right)$.

Profits from the safe good, conditional on stocking q_{st}, are $\pi_{st} = q_{st} p_s(q_{st}, \nu_{st}) - \zeta_s(q_{st})$ where $\zeta_s(\cdot)$ is a known and differentiable function capturing non-stock costs of selling, such as labor. Flow profits from the new good are determined by $\pi_{nt} = q_{nt} p_n(q_{nt}, \nu_{nt} + \theta) - \zeta_n(q_{nt})$ where $\zeta_n(\cdot)$ is also known and differentiable. I assume that profits are concave in quantity for each demand realization. Agents may invest in any non-negative stock of the safe product each period, $I_{st} \geq 0$ at wholesale cost w_s per unit. There is a minimum order size χ for the new product, so $I_{nt} \in \{0\} \cup [\chi, \infty)$.[3] The new product has a wholesale price of w_n. Since inventory is sold each period, $q_{jt} = I_{jt-1}$.

In addition to uncertain demand, agents face supply uncertainty when stocking the new product, capturing concerns that new suppliers may prove unreliable or shift focus and cut firms off from

[3] This condition captures realistic features of the market and allows for an equilibrium in which the agent does not fully learn demand. Imposing fixed delivery costs without a minimum order size yields similar results.

future supply. The model represents this as an exogenous probability Γ that the agent loses access to the new product beginning in period $t_c > 1$, which lowers the continuation value of learning.

Learning about demand: The agent learns about new product demand from stocking. They begin with a prior $\theta_1 \sim \mathcal{N}(\mu_1, \Sigma_1)$ and Bayesian update when their information set, $\mathcal{I}_t(I_{n0}, \ldots, I_{nt-1})$, changes. If the agent stocks I_{nt}, then in period $t+1$ they receive a signal $x(I_{nt}) = \theta_0 + \nu_{nt+1} + \eta_{nt+1}$, where $\eta_{nt+1} \sim \mathcal{N}(0, I_{nt}^{-1}\Sigma_x)$ is independent of demand and of other signals, so that $x(I_{nt}) \sim \mathcal{N}\left(\theta_0, I_{nt}^{-1}\Sigma_x + \Sigma_n\right)$.[4] The agent knows the signal precision and that it is centered on the truth, but not θ_0. Precision is increasing in I_{nt}, since more stock provides more opportunities for customer interactions. Learning is harder the more demand fluctuates, captured by Σ_n.

Agents can also learn from external sources, such as neighbors, who may reveal signals $x_{ot} \sim \mathcal{N}(\theta_0, \Sigma_o)$. I treat these signals as unanticipated: agents update their beliefs by Bayes' Rule when a signal arrives, but do not expect future signals when planning. Treating external signals as unanticipated rules out strategic free riding. While strategic considerations may be important, they are unlikely to change the tested predictions. This learning structure yields posterior beliefs

$$\begin{aligned} \theta &\sim \mathcal{N}(\mu_t, \Sigma_t) \\ \mu_t &= \Sigma_t \left(\Sigma_1^{-1}\mu_1 + \sum_{\tau: I_{n\tau}>0} (I_{n\tau}^{-1}\Sigma_x + \Sigma_n)^{-1} x(I_{n\tau}) + \sum_{x_{ot} \in \mathcal{I}_t} \Sigma_o^{-1} x_{ot} \right) \\ \Sigma_t &= \left(\Sigma_1^{-1} + \sum_{\tau: I_{n\tau}>0} (I_{n\tau}^{-1}\Sigma_x + \Sigma_n)^{-1} + |x_{o\tau} \in \mathcal{I}_t| \Sigma_o^{-1} \right)^{-1} \end{aligned} \tag{1}$$

The agent's objective: The entrepreneur receives flow utility of consumption given by a continuously differentiable, increasing, and weakly concave function $u(\cdot)$. They may save or borrow at interest rate r and discount the future at rate $\delta = \frac{1}{1+r}$. I model a borrowing limit $\underline{a} \leq 0$, capturing possible capital constraints. The agent begins with assets $a_0 > 0$ but no stock. Their objective is

$$\max_{\{c_t, a_t, I_{st}, I_{nt}\}} \mathbb{E}_0 \sum_{t=1}^{\infty} \delta^{t-1} u(c_t) \tag{2}$$

subject to a budget constraint $a_t + c_t + w_s I_{st} + w_n I_{nt} \leq (1+r)a_{t-1} + \pi_s(I_{st-1}, \nu_{st}) + \pi_n(I_{nt-1}, \nu_{nt} + \theta_0)$,

[4] Agents do not perfectly observe prices. This captures the fact that bargaining over prices is common, which adds noise to price realizations but also creates opportunities to learn from customer interactions, captured by $I_{nt}^{-1}\Sigma_x$.

minimum order sizes of the new product $I_{nt} = 0$ or $I_{nt} \geq \chi$, $I_{nt} = 0$ for $t \geq t_c$ if the agent has lost access, the borrowing limit $a_t \geq \underline{a}$, $I_{st} \geq 0$, and a transversality condition. Expectations are over θ, due to incomplete information, and ν_{jt}, due to stochastic variation in demand.

The value of learning affects new product stocking because investment facilitates learning, which improves future optimization. This enters the agent's objective as reductions in future "regret," lost utility due to incomplete knowledge of θ. Let $y_t = (1 + r)a_{t-1} + \pi_s(I_{st-1}, \nu_{st}) + \pi_n(I_{nt-1}, \nu_{nt} + \theta_0)$ be the agent's cash on hand. Define the conditional value function

$$V^*(y_t, \Gamma, \theta) = \max_{\{c_\tau, a_\tau, I_{s\tau}, I_{n\tau}\}} \sum_{\tau=t}^{\infty} \delta^{\tau-t} \mathbb{E}[u(c_\tau)|\theta] \tag{3}$$

subject to the same conditions as Equation 2, but treating θ as known. Let $\bar{c}_\tau$ denote consumption along the path that solves Equation 2, maximizing over beliefs instead of treating θ as known. Define

$$V(y_t, \mathcal{I}_t, \Gamma, \theta) = \sum_{\tau=t}^{\infty} \delta^{\tau-t} \mathbb{E}[u(\bar{c}_\tau)|\theta] \tag{4}$$

This is the expected utility the agent will receive from their planned actions if $\theta_0 = \theta$.

Regret is given by $R(y_t, \mathcal{I}_t, \Gamma, \theta) \equiv V^*(y_t, \Gamma, \theta) - V(y_t, \mathcal{I}_t, \Gamma, \theta) \geq 0$. This captures lost utility due to incomplete information if $\theta_0 = \theta$. Bayesian regret, $\bar{R}(y_t, \mathcal{I}_t, \Gamma) \equiv \mathbb{E}_\theta [R(y_t, \mathcal{I}_t, \Gamma, \theta)|\mathcal{I}_t]$, is the lifetime utility that the agent expects to lose because of uncertainty about θ.

2.2 Model solution

The model solution is derived in appendix A.1. I present and interpret the results in this section.

Optimal safe good investment: The optimal level of investment in the safe good, I^*_{st}, satisfies

$$\begin{aligned} &\delta \left\{ \mathbb{E}_t \left[\frac{u'(c_{t+1})}{u'(c_t)} \right] \mathbb{E}_t \left[\frac{\partial}{\partial I_{st}} \pi_s(I^*_{st}, \nu_{st+1}) \right] \right. \\ &\left. + \frac{1}{u'(c_t)} Cov_t \left(u'(c_{t+1}), \frac{\partial \pi_s(I^*_{st}, \nu_{st+1})}{\partial I_{st}} \right) \right\} = w_s - \frac{1}{u'(c_t)} \iota_{st} \end{aligned} \tag{5}$$

where $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot|\mathcal{I}_t]$ and $Cov_t(\cdot) = Cov(\cdot|\mathcal{I}_t)$. $\mathbb{E}_t \left[\frac{u'(c_{t+1})}{u'(c_t)} \right] = 1$ if capital constraints do not bind. Investment falls if one hits their borrowing limit, in which case $u'(c_t) > \mathbb{E}_t[u'(c_{t+1})]$ so the shadow cost of investment increases. ι_{st} is a Lagrangian multiplier ensuring non-negative investment.

$Cov_t \left(u'(c_{t+1}), \frac{\partial \pi_s(I^*_{st}, \nu_{st+1})}{\partial I_{st}} \right)$ captures possible risk aversion and will be zero if agents are risk

neutral (so $u'(\cdot)$ is constant). Low profits reduce consumption (if insurance markets are incomplete). If $u(\cdot)$ is concave, this covariance will be negative, leading to inefficiently low stocking levels.

Optimal investment in the new good: I^*_{nt} is defined by the Euler equation

$$\delta\Bigg\{\underbrace{\mathbb{E}_t\left[\frac{u'(c_{t+1})}{u'(c_t)}\right]}_{\text{Capital constraints}}\underbrace{\mathbb{E}_t\left[\frac{\partial}{\partial I_{nt}}\pi_n(I^*_{nt},\nu_{nt+1}+\theta)\right]}_{\text{Expected marginal profits}}+\underbrace{\frac{1}{u'(c_t)}Cov_t\left(u'(c_{t+1}),\frac{\partial\pi_n(I^*_{nt},\nu_{nt+1}+\theta)}{\partial I_{nt}}\right)}_{\text{Risk aversion}}$$
$$\underbrace{-\frac{1}{u'(c_t)}\frac{\partial}{\partial I_{nt}}\mathbb{E}_t\left[\bar{R}(y_{t+1},\mathcal{I}_{t+1}(I_{nt}),\Gamma)\right]}_{\text{Marginal learning value}}\Bigg\}=\underbrace{w_n-\frac{1}{u'(c_t)}\kappa_{\chi t}(2I_{nt}-\chi)}_{\text{Marginal investment costs}} \tag{6}$$

given that this exceeds the utility of not stocking. The marginal learning value term differentiates expected regret through the information set, holding cash on hand y_{t+1} fixed, since the effect of investment on liquidity is captured by expected marginal profits.

There are two important differences from Equation 5. First, expected profits and the covariance between profits and marginal utility depend on beliefs about θ, which learning changes. For instance, if the agent receives a signal centered on their prior, expected profits will be unchanged but a risk averse agent will perceive less stocking risk, reducing this term in magnitude.

Second, learning has value because it reduces regret. Letting $R_\tau(\theta) = u(c^*_\tau(\theta)) - u(\bar{c}_\tau(\theta))$ denote period τ regret, Appendix A.2 shows

$$\frac{\partial}{\partial I_{nt}}\mathbb{E}_t\left[\bar{R}(y_{t+1},\mathcal{I}_{t+1},\Gamma)\right]=-\frac{1}{2}\sum_{\tau=t+1}^{\infty}\delta^{\tau-t-1}Cov_t\Big(\overbrace{R_\tau(\theta)}^{\substack{\text{Sensitivity of}\\ \text{regret to }\theta}},\ \overbrace{\eta^2_{nt+1}\Sigma_x^{-1}}^{\substack{\text{Reduction in uncertainty}\\ \text{from investment}}}\Big)\leq 0 \tag{7}$$

In words, the increase in future utility one obtains from stocking more of the new product is a function of how strongly utility varies with θ and the sensitivity of learning to investment. Agents will stock past the point where the immediate payoff is 0 because investment allows them to better optimize in the future. This is the "risk-reward" trade-off typical of bandit models.

A core insight of this model is that the value of learning typically rises with demand uncertainty, Σ_t, because this implies a chance that the product is very profitable. Thus, a risk neutral firm's willingness to stock a new product is increasing in demand uncertainty, making it likely that profitable products are discovered. For risk averse firms, the potential downside associated with uncertain payoffs creates disutility, constraining experimentation. If risk aversion is sufficiently high, this

relationship may reverse: greater uncertainty can prevent firms from investing.

Supply chain uncertainty, Γ, lowers the continuation value of learning without affecting short-run returns, so it may be used to test the "risk-reward" trade-off. Learning about demand is valuable because an agent can restock if they discover a profitable product. Losing access limits this upside, reducing the value of learning while leaving profits for $t < t_c$ unaffected.

2.3 Comparative static predictions

I derive four testable predictions from the model: three study how risk aversion and demand uncertainty affect new product adoption, and one considers learning from external sources.[5]

Definition 1 (Mean-preserving contraction) *A mean-preserving contraction of the profit function at the minimum order size, χ, is a one-period perturbation, $\pi_n^p(\cdot)$, such that*

1. $\mathbb{E}_t\left[\pi_n^p(\chi, \nu_{nt+1} + \theta)\right] = \mathbb{E}_t\left[\pi_n(\chi, \nu_{nt+1} + \theta)\right]$
2. $\int_{-\infty}^{x} F_p(\chi, y)dy \leq \int_{-\infty}^{x} F(\chi, y)dy \quad \forall x$, *and strictly for some* x. $F(I_{nt}, y)$ *denotes the probability that* $\pi_n(I_{nt}, \nu_{nt+1} + \theta) \leq y$, *and* F_p *is the same for the perturbed profit function.*
3. $\pi_n^p(I_{nt}, \nu_{nt+1} + \theta) = \pi_n(I_{nt}, \nu_{nt+1} + \theta) \quad \forall I_{nt} \neq \chi$

Expected profits are unaffected by a mean-preserving contraction, but the distribution is less spread out. Since expected profits are unchanged, this will not affect a risk neutral firm's decision. However, a risk averse agent's expected utility of stocking will increase.

Proposition 1 (Risk aversion) *A mean-preserving contraction leads a firm to stock $I_{nt}^* > 0$ that otherwise would not only if they are risk averse.*

This provides a test for firm risk aversion that is not confounded by capital constraints binding today. Because the contraction lasts for one period, it also does not change the continuation value of learning. Interpreting the result as reflecting owners' risk preferences relies in part on an assumption that new product losses are small enough that a risk neutral owner could feasibly cover them by reducing consumption out of other firm profits. Absent this, profit maximizing firms may behave

[5] I focus on intuition for the propositions and their implications in the text. Proofs are presented in Appendix A.3.

as if risk averse if losses force them to bear convex costs (e.g. bankruptcy). This assumption is motivated by the empirical setting: other firm profits always exceed losses from helmets, so firm owners could have chosen to cut back on consumption rather than firm inputs.

To test whether learning about demand affects adoption, I analyze two policies: a one-time return offer and a resolution of supply chain uncertainty. A return policy guarantees that $p_{nt+1} \geq w_n$, reducing downside risk and raising expected profits. Resolved supply chain uncertainty, Γ, raises the continuation value of learning. Both policies should therefore increase new product investment if demand uncertainty binds. A sharper implication comes from the interaction between the two policies, which helps distinguish learning from alternative explanations. If the return policy fully insures the firm against losses (and other stocking constraints are slack), any agent who believes the product could be profitable will stock it. A reduction in Γ should then have no additional (extensive margin) effect among firms for which demand uncertainty is the binding constraint to adoption.

A final and more direct test of learning concerns the persistence of a one-time return policy. If firms learn about demand, then receiving returns once should cause them to experiment and update beliefs, affecting their future stocking behavior.

Proposition 2 (Learning about demand) *If firms face uncertain demand for a new product that they can overcome through learning*

a.) New product stocking increases when the firm is offered returns.

b.) Resolving future supply chain uncertainty, Γ*, increases current stocking if agents do not face binding capital constraints.*

c.) If $\delta \approx 1$*, non-stock costs of selling are low, and capital constraints are not binding, then a reduction in supply chain uncertainty has no effect on* $\mathbb{I}(I^*_{nt} > 0)$ *if a firm has access to returns.*

d.) A one-time return offer will have persistent effects on new product stocking.

Suppose that a one-time returns policy increases future stocking. This is consistent with risk aversion deterring firms from adopting the new product when they are uncertain about demand. Under this explanation, returns should cause firms with uncertain priors to stock, who then overcome uncertainty through learning. An alternative explanation is that returns cause firms with overly pessimistic beliefs to try stocking helmets, who correct their expectations. The model predicts a different relationship between beliefs and product adoption under these mechanisms.

Proposition 3 (Learning mechanism) *Suppose that a one-time returns policy has a persistent positive effect on new product adoption.*

a.) If returns correct the pessimistic priors of risk neutral firms, each of the following should hold:

i.) Prior uncertainty about demand positively affects uptake absent returns.

ii.) Firms induced to stock by returns hold more pessimistic or more certain beliefs.

iii.) Experience selling the new product causes firms to positively update expectations.

b.) If returns cause sufficiently risk averse firms to stock and overcome demand uncertainty:

i.) Prior uncertainty about demand negatively affects uptake absent returns.

ii.) Returns cause firms with uncertain priors to stock.

iii.) Experience reduces uncertainty about demand.

If firms learn by stocking, a natural question is whether they can also learn from other sources such as peers. Where information spillovers exist, a signal from neighbors should raise a risk averse firm's propensity to stock by lowering uncertainty, lowering the effects of the treatments. Formally,

Proposition 4 (Information externalities) *If firms learn about demand from neighbors*

a.) Conditional on expected demand, learning from peers lowers the value of returns, and the effect of reducing Γ *on the value of stocking falls as* Σ_t *falls to 0. Furthermore, the directional effects of learning from peers on stocking should match those of learning from one's own experience.*

b.) Peer signals increase I^*_{nt} *in expectation if agents are sufficiently risk averse or pessimistic.*

c.) Receipt of a signal x_{ot} *below an agent's expectation increases* I^*_{nt} *only if agents are risk averse.*

3 Setting and design of the experiments

I test the model's predictions using two field experiments in Kenya. First, an Insurance Experiment tests if firms are risk averse. Second, a Learning Experiment traces out the effects of risk aversion on learning by temporarily lowering stocking risk, then examining the persistence of effects.

Both experiments offer a motorcycle helmet introduced 2.5 years before the study began. Previously, helmets affordable to a typical Kenyan household offered minimal safety benefits. A motorcycle importer invested in a modern helmet factory in Kenya in 2020. Lower transportation and labor costs brought more effective helmets into the budget set of consumers.

Three features of the market make it well suited to studying new product adoption. First, helmets diffused slowly through retail markets: under 3% of screened shops stocked them at baseline. Second, motorcycle use, and therefore potential demand, is widespread. In 2022, an estimated 22 million motorcycle trips occurred each day.[6] Third, shops in the Learning Experiment sample could have stocked from the manufacturer for over two years, limiting access barriers to adoption.

3.1 The insurance experiment

Setting and timeline: The Insurance Experiment (November 2024 to April 2025) was conducted in 140 markets in West Kenya. The experiment included a baseline survey and a follow-up two months later, after which a separate *spillover* sample was surveyed.

The helmets studied were not widely available in West Kenya at the time of the experiment. Low-quality imports and other helmets used by motorcycle riders (e.g. construction helmets) were more common: about half of shops reported knowing of a shop selling such products at baseline.

Recruitment and sample: The baseline survey included 70 randomly selected markets, stratified by county. The remaining 70 markets were pure controls visited only for the *spillover* survey.

In each baseline market, field officers identified 10 shops suitable for helmet sales, prioritizing motorcycle spare part shops. If fewer than 10 existed, other shops such as hardware stores were added. Eligibility required that shops did not currently sell helmets and had a fixed physical location. From the 10 eligible shops, 5 were randomly selected, yielding 350 firms.[7]

A focus on large markets, and the requirement that enterprises have a permanent location, produced a sample of relatively large and established firms. Median profits in the month before the survey were about $325.[8] The average shopkeeper was 34 years old, had 13 years of education, and about 30% were female. Businesses had operated for over four years on average, and a quarter had an employee. Summary statistics are reported in Appendix Table A1.

The 5 shops not surveyed at baseline were targeted for the *spillover* survey. The aim was to offer these shops and identically sampled firms in pure control markets a stock of helmets to test for information externalities. However, only shop names were recorded at baseline to limit shop-

[6] Fred Matiang'i, "The urgency of bodaboda reforms", *Nation.Africa*, 2022.

[7] One market was inadvertently visited in piloting and at baseline, resulting in some firms receiving a control and treatment offer. Analysis focuses on the 345 remaining firms, but results are similar including the full sample.

[8] Egger et al. (2022) report median annualized revenue of $240 among retail firms in rural Siaya County, Kenya.

surveyor interactions, making it challenging to accurately re-identify shops.

As a result, field officers instead targeted the 5 best shops to sell helmets in pure control markets. In baseline markets, they recruited the 5 best shops not already surveyed (including skipped firms).[9] Typically, all motorcycle parts shops in each market were recruited, and assignment to the *insurance* or *spillover* sample was determined randomly in baseline markets. Consequently, the sample of motorcycle-related shops should be comparable across baseline and pure control markets within the *spillover* survey. Because fewer motorcycle shops were available to survey in baseline markets (since some had already been visited), firms in these markets are generally less suitable to sell helmets. Appendix Table A2 supports this view, showing that *spillover* shops in baseline markets are less likely to sell motorcycle products, but shops are balanced across other dimensions.

Design and randomization: The Insurance Experiment was designed to test whether risk aversion affects firms' stocking decisions. I randomly assigned firms to an *insurance* or control arm, stratified by market. *Insurance* firms were offered helmet stock with the option of a contract that reduces downside risk without raising expected profits. Control firms were offered the same stock without this option. I test for risk aversion by examining if stocking is higher among *insurance* firms.

Both arms were offered a stock of 3 helmets after a baseline survey and informed that they could purchase 3 or more helmets at a follow-up, regardless of whether they previously stocked. The order size (about 55% of median weekly profits) was the manufacturer's minimum. Shops paid full price but received free delivery, and had two weeks to place an order, giving time to search for customers.

Insurance shops were offered firm-specific contracts calibrated using subjective beliefs about demand. To elicit beliefs, *insurance* and control firms were asked how many of 10 helmet-stocking shops would sell out by follow-up, and whether their own shop would earn more revenue than the average firm, with incentives for accuracy. Surveyors then reported the implied probability of selling out, and shopkeepers stated their own belief about this probability, so_i.

Control firms were promised an unconditional payment (in Kenyan shillings) at follow-up of

$$P_i = \begin{cases} 1000 \cdot (1 - so_i) + 25, & so_i < 0.8 \\ 225, & so_i \geq 0.8 \end{cases}$$

[9]Some shops stocking helmets prior to the study, which were ineligible, reported adopting helmets after the study began to obtain subsidized stock. Back checks were conducted after the delivery of subsidized stock, so misreporting incentives were removed. Firms that disclosed eligibility violations in back checks are excluded from analysis.

Insurance firms chose between P_i and a contract paying 1,000 if they stocked helmets and failed to sell out, and 0 otherwise. The contract pays more than P_i when helmet demand is low but has a lower expected payout. Shops were not told that so_i determined P_i, and beliefs were elicited before the contract was introduced. Appendix A.4 shows that setting $P_i' = 1000 \cdot (1 - so_i)$ would make the contract a mean-preserving contraction of profits. Since $P_i > P_i'$, the contract is strictly dominated under risk neutrality, so higher stocking among *insurance* firms implies risk aversion.

The design addresses several confounding explanations. First, *insurance* shops chose between a future transfer and insurance, so the offer did not affect capital constraints when stocking (Casaburi and Willis, 2018). Second, both *insurance* and control firms were informed of the contract and its random assignment, so that receipt did not send a signal about demand. Third, firms accepting the payout could not restock, removing incentives to forgo sales. This also lowers the expected value of insurance, strengthening the strict dominance relationship. This appears to be an important source of strict dominance empirically, as shops regularly complained that the condition was unfair ex ante and multiple shops declined the payout with 1 unsold helmet remaining to restock ex post. Finally, a mystery shopper visited 1 in 5 firms to audit compliance, which every firm passed.

The primary outcome is an indicator equal to 1 if a firm ordered helmets within two weeks. A secondary "immediate adoption" outcome measures orders within a day, before firms could mitigate risk by finding customers. Surveys also measured firm characteristics, demand beliefs, realized sales, and risk preferences via a lottery-choice game conducted at the follow-up (Strobl, 2022). This exercise was omitted from the baseline to ensure that payouts did not relax liquidity constraints.

Markets had 2 or 3 *insurance* firms with equal probability. Within markets, simple random assignment allocated *insurance* to firms. Appendix Table A1 tests balance across arms.[10] After two weeks, if no firm in a market had stocked, shops were unexpectedly approached again (in a random order) and offered a large discount to induce stocking to test for spillovers.

3.2 The learning experiment

Setting and timeline: The Learning Experiment (January-July 2024) was conducted near the helmet factory in Nairobi, Kenya. A census identified eligible firms and collected data for randomiza-

[10]Treated shops are less likely to know of a shop selling helmets. This has no significant effect on stocking in this experiment (most of the helmets sold were poor substitutes). Controlling for it (Table 1 columns 3–4) has little effect.

tion. Data collection included a baseline survey (February-March), a midline (May) and an endline (June-July). Chronologically, this exercise was conducted before the Insurance Experiment.

Recruitment and sample: The Learning Experiment included 929 retailers. Shops with prior experience selling helmets and those without a permanent building were excluded. Recruitment targeted areas with few existing helmet sellers, so few firms were sampled from the Central Business District. Finally, shops that were certain they would never stock helmets were omitted, to approximate the population of firms a producer might target when seeking retailers.

Firms were recruited through a census. Surveyors visited shops and recorded information for randomization. Of 1,152 eligible firms listed, the target sample was 950–1,000. Shops were placed in a primary or a replacement pool. Firms far from existing sellers were placed in the primary pool, and the remaining firms were randomly assigned to one of the two. Surveyors recruited from the primary pool first, drawing replacements if firms declined. The final sample comprises 929 firms, as 22 interested shops did not complete the baseline and 5 firms were determined to be ineligible and dropped. Treatments were introduced after baseline, so that they did not affect participation.

Appendix Table A1 presents summary statistics. Firms in the Learning sample were larger than those in West Kenya, with average monthly profits over $600, and a smaller share (37%) sold motorcycle parts. Respondent age, gender and education levels were similar across samples, as were firms' likelihood of having an employee or adopting a new product in the prior year.

Returns treatment: The first experimental arm tested whether temporarily reducing inventory risk led shops to keep stocking helmets after the offer ended (Proposition 2). It was carried out in two phases. In phase one, all shops were offered helmet stock at market prices. *Returns* shops could return unsold stock from their first order for a full refund, whereas control shops could not. In phase two, both groups could stock without returns. I focus on whether *returns* raised phase two stocking by resolving demand uncertainty. As in the Insurance Experiment, all firms were informed of the treatments and their random assignment, so receiving an offer carried no signal about demand.

I use returns to reduce risk for two reasons. First, return policies are common in high-income settings (Padmanabhan and Png, 1995), but under 15% of *spillover* firms reported ever being made such an offer. The intervention therefore tests whether thin return markets constrain adoption. Second, unlike the insurance contract, *returns* involve no payouts that could affect phase two stocking.

A natural question is why the manufacturer did not offer returns. While this falls outside the

scope of what the study can rigorously address, the manufacturer's own account suggests that the structure of retail markets in LMICs may make return policies costly to provide. The company cited the contracting and transportation costs of accepting returns from many small retailers. It also believed a returns policy may be ineffective if offered by the supplier, since retailers mistrust new suppliers, whereas the study possessed national and local research permits.

During phase one, there was a minimum order size of 5 helmets, half the manufacturer's minimum at that time. This lowered liquidity needs among all shops and helped sustain relationships with control firms. Phase two ran from midline to endline, with orders accepted anytime. During midline, the manufacturer unexpectedly cut its minimum order size to 3, so the study matched this policy. The manufacturer did not indicate whether the policy shift was motivated by the study. However, the change took place shortly after the manufacturer observed the volume of orders the study received. The producer also asked the study team to share areas where shops had stocked.

If a shop indicated intent to order, surveyors revealed an installment option: firms could pay for their first order in three weekly payments, regardless of treatment. This helped shops with limited liquidity while preserving time for learning about demand. To prevent contamination, surveyors disclosed this option only after a shop expressed intent to buy. The default rate was under 3%.[11]

Shops could restock at any time during the Learning Experiment by calling to place an order, which was typically delivered within 1-2 days.

Supplier commitment treatment: The second treatment was designed to increase the continuation value of learning without affecting phase one helmet profitability or risk (Proposition 2). It did so by reducing uncertainty about future supply. Surveyors told *supplier commitment* shops that the study would help them restock directly with the manufacturer after study stocking ended, securing continued supply if helmets proved profitable. Control firms had to establish this link themselves. The design reflected concerns raised in piloting that the producer might change its policies or priorities, cutting small firms off from reliable supply.

A secondary information treatment: The endline survey included a randomized *information* treatment designed to test if firms learn from peers (Proposition 4). *Information* shops received data from 5 randomly selected phase one helmet adopters about how many helmets they sold, the

[11] In three cases, the option was revealed early; those firms were dropped. 91% of shops paid on time and 97% paid before the midline survey.

prices that they charged, and whether they restocked. Using 5 shops generated randomly varying signals: average sales were 3.8 helmets with a standard deviation of 1.2.

Randomization: Stratified random assignment was used to allocate the *returns* and *supplier commitment* treatments. I followed a 2x2 randomization design so that 1/4 of shops received *both* treatments. Randomization was stratified on neighborhood (for power), an indicator for whether the enterprise reported having any employees during the census (for power), and distance to the nearest existing helmet seller (for heterogeneity analysis). Appendix Table A1 verifies balance across arms.

The *information* treatment was delivered unexpectedly to half of shops that had not stocked helmets by the endline. Randomization was stratified on knowledge of a nearby helmet seller.

3.3 Primary variables collected

The baseline survey collected shopkeeper demographics and enterprise costs, revenues, and profits. Surveyors also elicited beliefs about helmet demand using a frequentist approach (Delavande et al., 2011). Respondents stated the price at which they would sell helmets, then were asked to imagine stocking 10 units. They were given 20 beans and a sheet with boxes labeled 0–10, and asked to distribute the beans to represent the probability of selling each quantity over a month.[12] I compute the mean and variance of each shop's belief distribution.

Reported expectations correlate strongly with responses to a direct question about anticipated sales ($t > 18, R^2 = 0.49$). Most respondents also provided plausible full distributions. A minority placed nearly all beans on a single value, yielding degenerate distributions. Results are stronger when these cases are excluded. Variance measures are also censored for shops with very high expectations, since the boxes stop at 10. Because the grid extends to twice the Learning Experiment's order size of 5, censoring binds only for shops expecting to sell well above the order size, which are unlikely to be at the margin of the *returns* offer.

The midline and endline surveys recorded helmet stocking from the study and other suppliers. For adopters, they captured helmet sales, costs, revenues, and profits, as well as future stocking intent. Updated firm information and beliefs about demand were collected from all enterprises.

I tracked helmet stocking decisions for all shops, including those that did not complete the

[12] On the first day of data collection, surveyors often entered totals that did not sum to 20. They were retrained, and the variable is set to missing for day one observations.

follow-up surveys. Most survey attrition comes from shops that did not wish to complete a survey but were willing to disclose if they had decided to stock or not. The remaining cases primarily reflect shops that closed and so were not stocking helmets at the time of the survey.

Analysis focuses on stocking decisions and on beliefs about demand. I also examine realized profits, but the experiment was not powered to detect profit effects since many shops never stock. I therefore emphasize repeat stocking over realized profits, since it provides revealed-preference evidence that stocking helmets maximizes utility for the firms that restock.

4 Empirical specifications

I examine the effects of the Insurance Experiment by estimating the regression

$$Stocked_i = \alpha + \beta Insurance_i + X_i'\gamma + \mu_m + \epsilon_i \tag{8}$$

where $Stocked_i$ indicates that shop i stocked helmets, $Insurance_i$ indicates assignment to *insurance*, X_i denotes controls, and μ_m are market fixed effects. Proposition 1 predicts $\beta > 0$, a positive investment response to *insurance*, only if firms are risk averse.

Turning to the Learning Experiment, I test if agents face demand uncertainty by estimating

$$Stocked_i = \alpha + \beta_1 R_i + \beta_2 S_i + \beta_3 R_i \times S_i + X_i'\gamma + \mu_k + \epsilon_i \tag{9}$$

$Stocked_i$ denotes phase one stocking, R_i indicates *returns* assignment, S_i the *supplier commitment*, and μ_k are strata fixed effects. Proposition 2 predicts (a) $\beta_1 > 0$, since *returns* reduce loss risk; (b) $\beta_2 > 0$, since the *supplier commitment* increases the continuation value of learning; and (c) $\beta_3 < 0$, since the *supplier commitment* should have little additional effect if *returns* eliminate risk. Part (d) predicts that *returns* will have persistent effects due to learning. I test this by replacing the outcome with phase two stocking and testing if $\beta_1 = 0$.

If returns increase phase two stocking, Proposition 3 differentiates between a case where *returns* overcome risk aversion versus incorrect beliefs. This is tested with three equations. First, I estimate

$$\begin{aligned} Stocked_i = \alpha &+ \beta_1 \mathbb{E}[Sales]_i + \beta_2 SD[Sales]_i + \beta_3 R_i \\ &+ \beta_4 R_i \times \mathbb{E}[Sales]_i + \beta_5 R_i \times SD[Sales]_i + X_i'\gamma + \epsilon_i \end{aligned} \tag{10}$$

$\mathbb{E}[Sales]_i$ is the agent's prior expectation about helmet sales next month and $SD[Sales]_i$ is the standard deviation of their belief. Proposition 3 predicts $\beta_2 < 0$ if risk aversion binds, meaning

control firms with more uncertain beliefs are less likely to stock. In contrast, Proposition 3 predicts $\beta_2 > 0$ if firms are risk neutral, reflecting learning value.[13]

Second, I examine the composition of adopters by estimating

$$Returns_i = \alpha + \beta_1 \mathbb{E}[Sales_i] + \beta_2 SD[Sales]_i + X_i'\gamma + \epsilon_i \tag{11}$$

among phase one adopters not receiving the *supplier commitment*. Under risk neutrality, $\beta_1 < 0$ or $\beta_2 < 0$, since *returns* should draw in firms with more pessimistic priors or lower learning value. If firms are risk averse, one may have $\beta_1 = 0$ and $\beta_2 > 0$, meaning *returns* induced firms with uncertain beliefs to adopt.

To examine how stocking affected beliefs, I estimate the two-stage least squares system

$$\Delta asinh(\mathbb{E}[Sales]_i) = \alpha_e + \beta_1 Stocked_i + \rho_e KS_i + X_i'\gamma_e + \mu_k + \epsilon_{ei} \tag{12a}$$

$$\Delta asinh(V[Sales]_i) = \alpha_s + \beta_2 Stocked_i + \rho_s KS_i + X_i'\gamma_s + \mu_k + \epsilon_{si} \tag{12b}$$

$$\begin{aligned} Stocked_i = {} & \pi_0 + \pi_1 R_i + \pi_2 S_i + \pi_3 R_i \times S_i + \pi_4 KS_i + \pi_5 R_i \times KS_i \\ & + \pi_6 S_i \times KS_i + \pi_7 R_i \times S_i \times KS_i + \mu_k + X_i'\gamma_1 + \nu_i \end{aligned} \tag{12c}$$

$Stocked_i$ denotes phase one stocking. I control for knowing a seller at baseline, KS_i, and use interactions between KS_i and treatment as instruments for power. $\Delta asinh(\mathbb{E}[Sales]_i)$ is the change in the inverse hyperbolic sine of i's expected sales from baseline. $\Delta asinh(V[Sales]_i)$ is the analogous change in i's variance of beliefs. Proposition 3 predicts $\beta_1 > 0$ if experience corrects pessimistic priors. It predicts that $\beta_2 < 0$ if experience mitigates risk aversion by reducing demand uncertainty.

The final class of predictions addresses spillovers. Let $TM_i = \mathbb{I}$(Baseline *insurance* market). IT_i denotes receipt of the *information* treatment and HS_i is one if the average sales in the *information* signal is greater than an agent's expectation. I test for information externalities by estimating

$$\begin{aligned} Stocked_i = {} & \alpha_1 + \beta_1 R_i + \beta_2 S_i + \beta_3 R_i \times S_i + \beta_4 KS_i + \beta_5 KS_i \times R_i \\ & + \beta_6 KS_i \times S_i + \beta_7 KS_i \times R_i \times S_i + X_i'\gamma_1 + \epsilon_{1i} \end{aligned} \tag{13a}$$

$$Stocked_i = \alpha_2 + \beta_8 TM_i + \mu_c + \epsilon_{2i} \tag{13b}$$

$$Stocked_i = \alpha_3 + \beta_9 IT_i + \beta_{10} HS_i + X_i'\gamma_2 + \mu_n + \epsilon_{3i} \tag{13c}$$

[13] I estimate this model among the sample of shops not receiving the supplier commitment since the focus is on untreated behavior and the effect of returns.

Proposition 4 part (a) predicts that learning from incumbents lowers the value of *returns* and the *supplier commitment*, implying $\beta_5 < 0$ and $\beta_6 < 0$. Part (b) states that externalities exist if exposure to a seller increases adoption, implying $\beta_8 > 0$. Finally, Equation 13c tests whether spillovers operate through expectations or uncertainty. If they correct pessimistic priors, a signal only raises stocking when it exceeds expectations ($\beta_9 < 0$, $\beta_{10} > 0$). If they mitigate uncertainty for risk averse firms, then even a slightly negative signal can increase stocking, implying $\beta_9 > 0$.

5 Results

Descriptive evidence from firms' beliefs supports the view that stocking helmets is viewed as a profitable but risky investment (Appendix Table A3). In the Learning Experiment, where helmets were offered at market prices, the median firm perceived a 50% chance that stocking helmets would increase firm profits versus a 30% chance that it would lead to losses. Motivated by this, I turn to the Insurance Experiment to examine whether risk aversion constrains stocking.

5.1 Firm risk aversion

The results of the Insurance Experiment (Equation 8) show that access to the insurance contract significantly increased helmet adoption, implying a rejection of firm risk neutrality.

Insurance increased the rate of shops that acquired stock within 24 hours by 7 percentage points on a base of 5% (Table 1 Panel A, $p < .01$). Including shops that stocked later, the effect is about 10 percentage points (50%, $p = .02$). This shows that lowering profit variance (while reducing expected returns) substantially increases the propensity of firms to stock a new product, implying that risk aversion affects firms' investment decisions. Effects of *insurance* are more than twice as large among firms operated by more risk averse shopkeepers, consistent with a lack of separability between owners' consumption preferences and production decisions ($p = .01$, column 6).

Shopkeeper risk aversion was measured at follow-up so that payouts from the game did not relax credit constraints over the period where *insurance* adoption effects are measured. But this could bias the analysis if the treatment affected measured risk aversion. There is little evidence of this in the data. *Insurance* has no detectable effect on the estimated CRRA parameter (Appendix Table A4), and treatment explains essentially none of the variation in risk preferences (within $R^2 = 0.001$)

or a firm owner's propensity to have above-median risk aversion. Evidence of heterogeneity in *insurance* effects by risk preferences also exists using the 75th percentile to classify firm owners and using the continuous measure, indicating that results are not driven by the cutoff used.

The rate and composition of *insurance* shops that opted into the contract suggest high comprehension of the offer. Half of *insurance* shops that immediately stocked selected the contract, and 40% of firms that ever stocked did (Table 1 Panel B). These shares align with treatment effects: the immediate effect implies roughly 13 additional adopters, and 11 shops took the contract, consistent with the contract being pivotal for the marginal firm. Shops with more uncertain beliefs were more likely to choose *insurance* (bootstrapped $p = .05$, column 2). That shops with little demand uncertainty typically declined *insurance* is consistent with their understanding that the contract offered a lower expected payout. Anecdotally, shops often complained that *insurance* should be more generous, objecting to the fact that firms collecting the payout could not restock.

A potential concern is that *insurance* could raise expected profits if beliefs were mismeasured. I focus on stocking effects of *insurance* (not whether shops accept the contract) because small measurement error is unlikely to make stocking profitable. Four tests also suggest that mismeasured beliefs do not drive the results. First, firms that selected *insurance* would on average have received higher payouts had they declined the contract, meaning the contract is also dominated under rational expectations. Second, if mismeasurement drove the results, take-up should be higher where the gap between the contract's expected payout under elicited beliefs and the guaranteed payment was smaller, since less measurement error would then be needed to reverse which option maximizes the expected payout. No such relationship appears in the data, although standard errors are wide (Table 1, Panel B). Third, elicited beliefs are informative: shops' stated probability of selling out predicts the realization, with an elasticity of 0.2 among all helmet adopters ($p = .05$) and 0.4 among those that accepted *insurance*. Fourth, I recomputed expectations using the frequentist data. On average, the expected value of *insurance* falls. However, the frequentist data are noisy, so dominance only holds for 70% of firms (a lower bound since this calculation ignores the restocking condition). Restricting the sample to firms for which the contract is dominated under both measures yields similar *insurance* effects: 6.2 percentage points on stocking within 24 hours ($p = .08$, Table 1, column 7) and 11.2 percentage points within two weeks ($p = .04$, column 8).

Treatment effects remain large and significant among larger and older firms. *Insurance* in-

creased immediate helmet adoption by 20 percentage points ($p < .01$) among firms with employees and the rate of shops that ever stocked helmets by 25 percentage points (102%, $p = .01$, Table 2). Point estimates are also large and significant among older firms and those with higher profits.[14] These results suggest that risk aversion is a barrier to firm investment even when enterprises have liquidity available to invest. The fact that larger firms suggestively respond more to insurance (differences between small and large firms are generally not significant) likely reflects the fact that larger firms are less likely to face binding capital constraints, not greater risk aversion among bigger enterprises. Correlations between owners' lottery choice measured risk aversion and firm size are negative, a statistically significant relationship for business age ($p = .02$).

The larger response to *insurance* among firms with more risk averse owners is consistent with effects operating through owners' preferences. An alternative mechanism is that if stocking helmets crowded out other inventory, convex borrowing costs could lead risk neutral firms to favor safer investments. That explanation predicts larger *insurance* effects among smaller firms, the opposite of what is observed. Nor does it predict that *insurance* effects should vary with owners' risk preferences. Directly supporting the view that such constraints are slack among compliers, under 4% of shops that adopted helmets reported having to stock less of other goods to afford them (shops that stocked reported more free space). Finally, the effects of *insurance* remain significant, and are suggestively larger, among firms reporting above-median baseline access to liquidity.[15]

The aim of this paper is to produce a revealed-preference test of firm risk aversion and examine its implications for product adoption. This provides direct evidence that risk aversion matters for firm investment with few structural assumptions. But these results raise the question of how much risk aversion is needed to rationalize firms' responses to *insurance*. Appendix B considers this question using (1) a simple model of the stocking decision of firms with CRRA utility, identified using random assignment to *insurance*; and (2) back-of-the-envelope calculations of the risk aversion needed to rationalize firms' decisions to opt into insurance. These results are speculative. The experiments were not designed to generate the variation needed to precisely estimate utility curvature, and weak-instrument-robust confidence sets are uninformative as a result. However, the

[14] Control stocking rates do not differ substantially by firm size. This likely reflects shops that did not perceive risk from stocking and those that found customers before stocking.

[15] The coefficient on *insurance* × above-median liquidity is 0.20 ($p = .06$) in a regression of uptake on *insurance*, high liquidity, their interaction, and controls for family income, beliefs about demand, and employee hours.

point estimates change in expected ways when background income and risk are considered, and they are higher among owners who were more risk averse in the lottery choice game. Both approaches suggest that the results can be rationalized with conventional levels of risk aversion, provided that firms cannot borrow to smooth consumption, although I cannot rule out larger CRRA values.

These calibrations should be interpreted in light of the fact that, while the cost of stocking and the value of *insurance* are modest in absolute terms, the stakes are high relative to household income, which for over 60% of the sample consists entirely of firm profits. Scaling these values by the ratio of income per person in the United States to that in this sample, using median US household income of $87,460 (2025 Census Bureau estimate) and an average household size of 2.5, against median profits last year of $2,603 and an average household size of 3.9, the stocking cost is equivalent to an expenditure of about $2,510 for a US household, with *insurance* paying out about $1,140. American households adjust consumption considerably in response to changes in income of this size (Parker et al., 2013). The results may therefore reflect that even modest risks represent a substantial share of income among the poor, rather than unusually high risk aversion.

5.2 Learning about demand

The effects of *returns* and the *supplier commitment* on stocking in phases one and two of the Learning Experiment (Equation 9) match the predictions of Proposition 2, consistent with demand uncertainty and learning consequentially affecting helmet stocking.

Returns increased the rate of firms that stock helmets during phase one by 9 percentage points relative to a 6.8% control rate (128%, $p < .01$, Table 3). The effect remains stable, 8 percentage points ($p = .02$), when the dependent variable is an indicator for ever stocking helmets by the endline survey. This suggests that *returns* increased firms' willingness to experiment with selling helmets, rather than simply accelerating adoption among firms that would have stocked regardless. That effects are modest in absolute terms likely reflects the fact that new product experimentation is rare: only 34% of shops reported experimenting with stocking any new product in the year preceding the survey. Absolute effects are also limited by the fact that the Learning Experiment did not target by industry. Among shops selling spare motorcycle parts, *returns* raised phase one stocking by 21 percentage points on a base of 15% ($p < .01$, Appendix Figure A1).[16]

[16]I included non-motorcycle shops since helmets could diversify product stocking for these firms. In practice, this

The *supplier commitment* increased phase one helmet stocking by 6 percentage points among firms not offered *returns* ($p = .03$, Table 3). The model predicts that it should induce no additional stocking once risk is removed, since shops constrained by uncertainty should already stock. Consistent with this, the effect falls to 1.8 percentage points among firms offered *returns* (−0.3 without controls). The difference in effects is not significant overall ($p = .32$). However, among firms that reported no knowledge of a helmet seller at baseline, the one pre-registered dimension of heterogeneity, the difference is larger and significant ($p = .08$ with controls, $p = .04$ without; Appendix Table A5).[17] This supports the interpretation that firms internalize a "risk-reward" trade-off, increasing investment when the future value of learning increases. This also weighs against learning by doing, which would predict a *supplier commitment* effect even when *returns* are offered.[18]

These predictions apply to firms for which demand uncertainty is the binding constraint, so the fact that a majority of shops did not stock even with *returns* does not contradict the model. While other explanations could rationalize the results, one possibility is that neither treatment was strong enough to induce adoption when barriers like capital constraints bind. Descriptive evidence supports this view. First, under 2% of helmet adopters stocked less of another product, suggesting slack capital constraints among compliers. Second, under 15% of shops in the Learning Experiment reported any fixed costs of selling helmets, averaging under $2.50 (Appendix Table A10). Third, shops did not typically report working longer hours to sell helmets. This interpretation also aligns with the Insurance Experiment, in which larger firms, if anything, responded more to *insurance*.

Returns had large and significant persistent effects on stocking in phase two, when *returns* and control shops faced identical stocking conditions. *Returns* firms were 7 percentage points (70%, $p = .02$) more likely to stock in this phase (Table 3). *Returns* also had large effects on market entry, defined by restocking and reporting intent to continue selling helmets. *Returns* effects doubled the number of permanent market entrants based on this definition (Table 3). These results suggest that agents learned about demand since the groups differ only in that *returns* shops had more experience selling helmets in phase one, matching Proposition 2.

The persistent effect of *returns* on stocking provides revealed preference evidence that many

seems to be second order relative to uncertainty about whether helmets will sell or not.

[17] While I did not file pre-analysis plans, I did pre-register primary outcomes and heterogeneity tests.

[18] Ceiling effects could in principle produce this pattern, but the *returns* arm does not appear saturated: 16.5% of *returns* firms stocked, and another 30.7% expressed interest without purchasing (Appendix Figure A1).

shops found helmets profitable, consistent with descriptive data. On average, phase one adopters reported profits of $70 from helmet sales by endline, with fewer than 7% reporting losses. Shops sold about 10 helmets on average, at a typical price 1.5 times the wholesale cost. Of phase one adopters, 65% restocked and over 80% reported intent to stay in the market (Figure 1). By endline, stocking shops reported that helmets accounted for about 10% of their total profits.[19] The decision is nonetheless risky: some shops reported losses exceeding $100. Consistent with these patterns, few *returns* shops (7% of adopters) returned any helmets, and returns were concentrated among loss-making firms: 80% of returning firms reported losses, versus 3% of shops that made no returns.

5.3 Does risk aversion prevent experimentation with uncertain new products?

These results suggest that *returns* helped many firms discover a profitable product. The findings are consistent with risk aversion inhibiting experimentation with products whose demand is uncertain, but could also be driven by *returns* increasing expected profits, correcting pessimistic expectations.

The results of Equations 10-12 indicate that *returns* caused risk averse firms to try stocking helmets, who overcame demand uncertainty through learning. Each prediction of Proposition 3 under this mechanism is satisfied, while there is no significant evidence for any of the conditions predicted under misperceptions. The results suggest that risk aversion can substantially inhibit new product adoption by deterring firms from experimenting with risky products.

First, the results of Equation 10 show that, absent *returns*, firms more uncertain about the profitability of helmets are significantly less likely to stock (Table 4). This is consistent with risk aversion undermining experimentation: firms uncertain about demand avoid stocking. Otherwise, one would expect the opposite, since the option value of learning rises with uncertainty. I estimate regressions of phase one stocking on $SD[Sales]_i$ plus this variable interacted with assignment to *returns*, controlling for expectations. As discussed in Section 3.3, $SD[Sales]_i$ is truncated when expected sales approach 10. Furthermore, if expectations are sufficiently high, $SD[Sales]_i$ is an imperfect proxy for uncertainty about the profitability of stocking 5 helmets.[20] I thus exclude cases where firms perceive no risk helmets will yield losses (29% of cases).

[19] Calculating this share by normalizing cumulative endline profits by months stocking (measured to the day) yields a value near 5%. Flooding and protests disrupted firms' business earlier in the study, which may explain this difference.

[20] Put differently, if a shop is uncertain about selling 5 helmets or 10, risk aversion should not prevent purchasing 5. So a censoring issue affects the binary outcome measure for high expectations.

Estimates in column 1 of Table 4, which includes rich firm and respondent controls, indicate that a one helmet increase in the standard deviation of beliefs about sales reduces firms' propensity to stock by 5 percentage points (85%, $p = .02$). This relationship is similar using beliefs about revenue rather than sales, excluding covariates, and selecting controls by double-post LASSO (Appendix Table A6). Estimates retain the same sign but are smaller and not statistically significant when observations prone to censoring are kept (−0.022, SE 0.017). An additional check exploits the fact that the outcome is a binary decision to stock 5 helmets. If a firm's beliefs are high enough that it perceives no risk of loss from acquiring 5 helmets, theory no longer predicts a negative relationship between stocking and the standard deviation of beliefs. Consistent with this, the standard deviation of sales positively predicts stocking among firms perceiving no loss risk (Appendix Table A6, column 6), although the difference from other firms is not statistically significant ($p = .24$).

Second, the results of Equation 11 indicate that firms that stocked with *returns* held more uncertain but similarly optimistic beliefs versus control adopters (Appendix Table A7). This is consistent with treatment compliers being constrained from stocking by demand uncertainty and risk aversion, not incorrect beliefs. Among phase one adopters, a one helmet increase in the standard deviation of baseline sales beliefs is associated with a 20 percentage point higher likelihood that the shop adopted with *returns* versus the control group (bootstrapped $p = .06$, $p = .04$ in a log specification), while expected sales are similar across *returns* and control adopters (−0.001, SE 0.036). This compositional effect is also evident in Table 4, which shows a positive interaction between treatment assignment and belief uncertainty (this result is not significant with linear controls for expectations, but is once non-linear controls for expected sales are added in column 3).

Third, and most directly, there is little evidence that the expectations of phase one adopters positively updated, but there is evidence of reduced demand uncertainty. I examine how experience selling helmets affected compliers' beliefs using Equation 12. The challenge is that a minority of shops stock helmets and beliefs are measured with noise, limiting power. I therefore interact treatment assignment with knowledge of a baseline seller (the one pre-registered dimension of heterogeneity) and control for this variable, which allows first stage predictions to vary by pre-existing seller knowledge while still identifying estimates entirely off of randomly assigned variation. This approach increases instrument efficiency by logic similar to Abadie et al. (2023).

Table 5 shows that there is little evidence that firms' expected sales positively updated with

experience. Point estimates are small, and slightly negative at endline, although confidence intervals cannot rule out positive updating. In contrast, estimates indicate that stocking caused the variance of compliers' beliefs about demand to fall sharply (weak IV robust $p = .03$).[21] Confidence intervals are wide, but the finding that uncertainty fell is robust to applying weak instrument robust inference (motivated by modest first stage F-statistics). Results are similar using levels instead of the inverse hyperbolic sine of beliefs about sales, and applying the same specification to beliefs about revenue rather than sales yields a small and insignificant point estimate on expected profits but suggests a \$26 decrease in the standard deviation of revenue (weak IV robust $p = .05$, Appendix Table A8).

The instrumental variable results efficiently combine information from both treatments to estimate a local average treatment effect of experience on compliers' beliefs, but estimates are at the edge of statistical power. Additional tests support the view that uncertainty fell while expectations did not change significantly. First, the reduced form effect of treatment assignment shows that *returns* reduced posterior uncertainty, though only marginally significantly ($p = .08$, Table 5), with a point estimate near zero for expected sales. Second, regressing the change in beliefs on phase one stocking without an instrument, essentially a difference-in-differences estimate, shows that stocking was associated with a 0.02 unit reduction in the inverse hyperbolic sine of expected sales and a 0.19 unit reduction in the variance of sales ($p = .04$, Appendix Table A8). Third, if agents become less uncertain about demand, their beliefs should better predict realized sales once they gain experience. The elasticity of midline sales with respect to baseline beliefs is 0.28 ($p = .02$, $R^2 = .045$; Appendix Table A9). Among shops that stocked in phase one, the elasticity of endline sales with respect to midline beliefs rises sharply to 0.69 ($p < .01$, $R^2 = .239$), and one can reject the null of no change in predictive accuracy with 95% confidence.

Finally, I directly test the relationship between belief updating and entry by estimating a regression of endline market entry on the change in (i) expected sales and (ii) the standard deviation of sales from baseline to midline among phase one adopters, controlling for bins of midline expectations (Appendix Table A8, Panel B). Under risk neutrality, shops whose uncertainty fell by less retain more learning value and so should be weakly more likely to enter, holding expectations fixed, implying a non-negative coefficient on $\Delta SD[\text{sales}]$. This holds whether priors were pessimistic or

[21] The pooled point estimate implies $\exp(-1.45) - 1 \approx -0.77$ relative to the untreated counterfactual, and adding the mean change among control firms (0.18) gives compliers' total change from baseline, $\exp(0.18 - 1.45) - 1 \approx -0.72$. Both treat the inverse hyperbolic sine as a log, an approximation at observed levels of $V[Sales]$.

correctly centered, so the test does not depend on how beliefs updated on average. Empirically, the opposite holds, as predicted if risk aversion deters shops uncertain about demand from stocking. A one helmet larger fall in the standard deviation of sales raises the probability of entry by 13.2 percentage points ($p = .03$), while changes in expected sales have little predictive power ($\beta = 0.02$) once midline expectations are controlled for.

Tests of learning therefore consistently fit predictions if *returns* induced risk averse firms constrained by uncertainty to stock, who perceived stocking as less risky once they learned about demand. Rationalizing the results through risk neutral firms' changing expectations would require a large mass of firms to grow more optimistic, as a majority of phase one adopters restocked, and lower posterior uncertainty reduces a risk neutral firm's incentive to stock. Yet expectations show no systematic increase, and 40% of entrants lowered their expected sales. Moreover, the negative correlation between uncertainty and control uptake, and the finding that larger reductions in uncertainty predict greater restocking, both have the opposite sign from what risk neutral learning predicts (although neither is experimentally identified). Evidence on how firms learned similarly indicates that experience reduced demand uncertainty, while point estimates, though imprecise, show no systematic change in expectations. Taken together, these results align with risk averse learning and indicate that risk neutral learning does not likely explain stocking persistence.

5.4 Learning from external sources

The Learning Experiment indicates that firms learn about demand by stocking new products. Estimates of Equation 13 indicate that firms can also learn from neighbors that stock, consistent with Proposition 4. This supports the view that the persistent *returns* effects reflect learning about demand, since information from sources other than a firm's own experience affects stocking as well.

I first examine treatment effect heterogeneity by exposure to a baseline seller in Table A5. Point estimates indicate that neither the *returns* nor the *supplier commitment* treatment affected stocking when there was a pre-existing helmet seller near the firm. The absence of effects in these markets reflects a greater propensity of control firms to stock: the control stocking rate in markets with a pre-existing seller (17%) approximately matches the *returns* rate in markets without one (16%). These results are somewhat underpowered, as only 27% of shops knew of a baseline seller, but one can reject equality of *returns* estimates when covariates are included ($p = .04$).

Results of the *spillover* survey support the view that firms learn from each other (Table 7). The coefficient on "BL market" reports the effect of being in a market where a shop was induced to stock helmets three months earlier. Results are reported over motorcycle-related shops and for all shops in the sample. As detailed in the design, the sample of motorcycle shops is likely balanced across markets, whereas in the full sample, firms in baseline markets were ex ante less likely to stock.[22]

Firms observed competitors adopt helmets and were more likely to stock themselves. Motorcycle shops were about 13 percentage points more likely to report knowing a helmet seller in their market ($p = .03$) and about 10 percentage points more likely to purchase stock ($p = .01$). Across all shops, the estimated effect on stocking is 6 percentage points ($p = .01$). A caveat is that baseline markets also experienced several months of research activity absent in pure control markets. This should not affect reported knowledge of a local seller, but could raise willingness to purchase from the study, so the stocking estimates may partly reflect familiarity with the research team. Anecdotal reports suggest that the effects more likely reflect spillovers. Firms reported that customers entered their shops with pictures of the study helmets and asked if they could provide them at a better price.

I test directly if information matters using the *information* treatment (Table 6). Receiving sales data increased the share of shops stocking helmets from 0.6% to 2.8% ($p = .02$). As with *returns*, the evidence suggests that information affected behavior by reducing uncertainty. When a "high signal" indicator—equal to 1 if the data a shop received exceeded its expectation—is added (Equation 13c), the effect of receiving any data is similar and significant. This interaction is based on a relatively small number of adopters so should be viewed cautiously, but aligns with evidence from firms' direct learning that learning may increase stocking even if it does not increase expectations.

Finally, evidence suggests that spillovers were localized, explaining slow helmet diffusion in Nairobi. In the *spillover* sample, only 42% of shops in treated markets reported knowing of a local helmet seller. In the Learning Experiment, shops within 0.25km (about 3 blocks) of a study shop that adopted helmets were 14 percentage points more likely to report knowing of a seller ($p < .01$), but the presence of a shop within 0.25-0.5 km had no detectable effect (Appendix Table A11). This suggests minimal spillovers across markets, preventing social learning from ensuring diffusion.

[22] Surveyed shops in baseline markets are 20 percentage points less likely to sell motorcycle parts, and in pure control markets, motorcycle-related shops are more than three times as likely to stock helmets as other shops.

6 Discussion

This section addresses whether the persistence of *returns* effects can be attributed to mechanisms other than learning that resolves demand uncertainty among risk averse firms. I also consider the implications of the results for the size of the retail market for new products.

I consider three alternatives. The first is that *returns* increased profits, relaxing liquidity constraints and increasing phase two stocking. But *returns* had no detectable effect on overall firm profits, as high profit volatility dominated helmet profits. Stocking effects also survived an unexpected drop in the minimum order size from 5 to 3, which lowered liquidity needs for all firms. Lastly, flooding at the end of phase one reduced profits by over 10% among affected shops ($p = .04$, Appendix Figure A2). If *returns* raised phase two stocking by relaxing liquidity constraints, such a shock should have eliminated the effect among these firms. Yet phase two stocking among *returns* shops hit by flooding remained higher than among control shops that were not flooded ($p = .02$).

A second possibility is that dynamics like learning by doing or customer capital could raise the profitability of selling helmets with tenure. The fact that firms' expected sales show little growth over time, combined with low reported non-stock costs, speaks against such forces. Furthermore, I consider the set of firms that stocked by the midline survey of the Learning Experiment, then regress midline to endline helmet profits on tenure. This leverages variation from the exact day shops began stocking (many waited for profits from other goods to stock). A concern is that shops that stocked earlier and discovered they were not profitable may exit, so I restrict the sample to shops that planned to stay in the market at endline. Estimates indicate that one additional month of tenure is associated with an insignificant decrease in profits, and confidence intervals reject values above $3 (Appendix Table A10). The lack of *supplier commitment* effects given *returns* is also inconsistent with such explanations. Finally, the finding that information about demand from external sources increases stocking provides positive evidence for the learning about demand explanation.

A third possibility is that behavioral biases such as loss aversion, narrow bracketing, or ambiguity aversion make insurance valuable for reasons beyond neoclassical risk aversion. The core argument of this paper is that small firms' production decisions depend on owners' risk preferences, violating strict profit maximization. This still holds, and may be more important, if behavioral biases affect risk preferences. For instance, risk aversion may affect decisions even as enterprises

become large if they narrow bracket. The evidence that plausible values of risk aversion can rationalize the findings (given difficulties smoothing consumption) suggests that appealing to behavioral explanations may not be needed to explain the results, though this does not rule them out. The response to the *supplier commitment* is also inconsistent with preferences where only the worst outcome matters (e.g. maxmin utility) since investment expanded when the future value of learning rose, with no change in downside risk. Future research designed to test if firms exhibit non-standard risk preferences would be valuable to further understand how firms respond to risk.

The expansion in helmet stocking generated by *returns* indicates that risk aversion can constrain the retail market for new products, which may harm the profitability of upstream producers. The extent to which retailer risk aversion harms upstream incentives depends on (1) whether the intervention expanded market access or displaced adoption that would have occurred anyway, and (2) whether treated firms would have eventually stocked helmets without the intervention.

The positive spillover results suggest that displacement is unlikely. Appendix Table A12 further shows that *returns* increased the likelihood that the shop or a nearby enterprise sold helmets at endline by about 8 percentage points (32%, $p = .04$). If *returns* merely shifted which shops stocked, they would have no effect on whether any seller operated nearby, so this measure is more robust to extensive margin displacement. It does not rule out that customers would otherwise have bought helmets from shops in other areas, though search costs make pure displacement of this kind unlikely.

Whether *returns* entrants would have eventually stocked without the treatment is harder to assess since the study ended after 6 months. However, helmets were available to shops for over two years before the study, suggesting that shops were unlikely to enter absent the study. Effects on helmet sales during the study were also important even if the study did not change long-run access. Appendix Table A12 indicates that *returns* induced shops to sell about 1.4 more helmets on average during the study (140%, $p = .07$). In markets with no baseline helmet seller, this jumps to 2.1 (517%, $p = .03$). This corresponds to *returns* inducing over 600 additional helmet sales in this time, accounting for around $15,000 in sales.

7 Conclusion

This paper provides evidence that risk aversion deters small retail firms in Kenya from adopting a profitable new product. Offering firms an insurance contract that is strictly dominated under risk

neutrality increased adoption by 50%, yielding a revealed-preference test of risk aversion that can be adapted to other settings. I then examine the dynamic consequences of risk aversion by testing a model of firm learning. A temporary return policy raised stocking by 70% after the offer ended. Evidence indicates that these persistent effects arose because *returns* induced risk averse firms to experiment, and those firms then resolved demand uncertainty through learning. Finally, addressing firms' concerns about the reliability of future supply raised the future value of information and expanded experimentation by 88%, as bandit models of learning predict.

The expansion in product stocking generated by these interventions suggests, speculatively, that risk aversion can constrain the retail market for new products in LMICs. Beyond slowing retailer growth, these distortions may harm incentives for upstream producers to introduce new goods or invest in context-specific innovations in the first place. While mechanisms like return policies can mitigate retailer risk aversion, the logistical, transportation, and contracting costs of these offers may be prohibitive when selling to many small retailers. Consequently, the prevalence of small retailers in developing markets may disadvantage innovative producers. Future research testing how producers respond to retailer risk aversion would be valuable to test these possible implications.

More broadly, the finding that risk aversion can have first-order effects on small firms' decisions has important implications for the interpretation of firm behavior and for policies designed to promote growth. Risk aversion could contribute to features of LMICs such as slow technology diffusion and high returns to capital. It may also lower the effectiveness of cash transfers if risk averse firms direct capital towards safe, low-return investments. Investigating the role of risk aversion in such choices, and testing policies to counteract it, could be valuable to better align academic understanding of LMIC firms with their behavior and to identify effective policies to promote growth.

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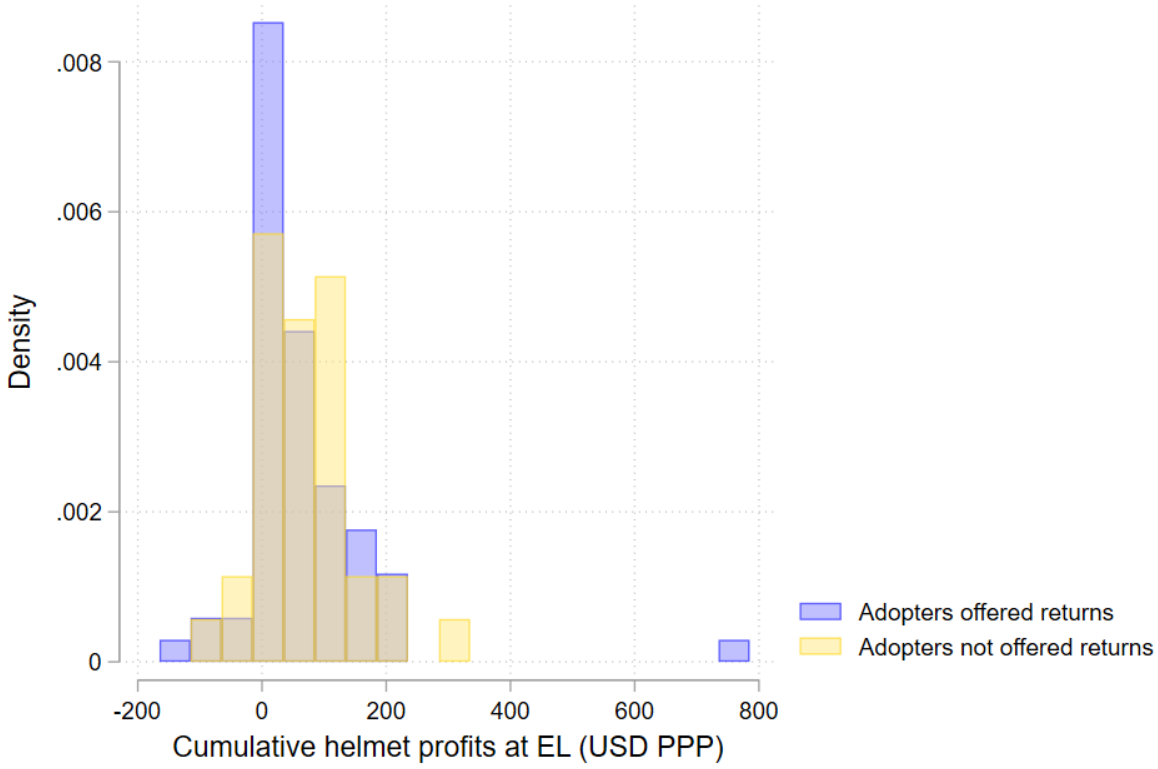
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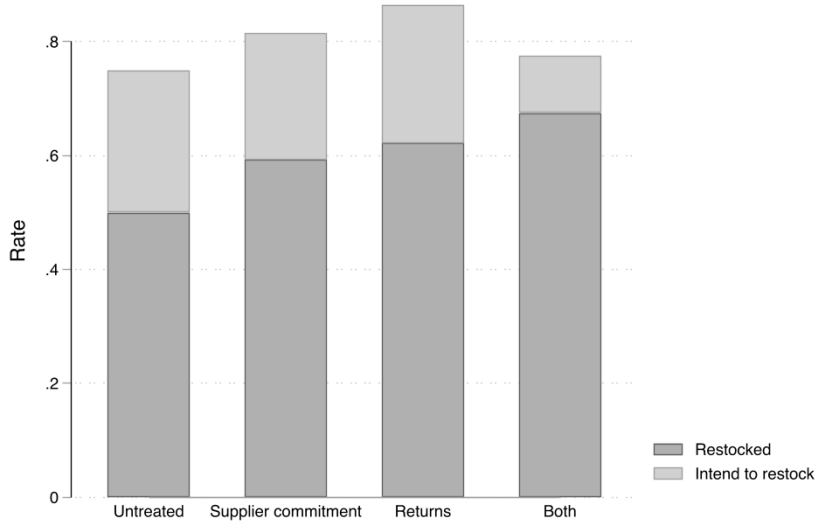
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Figure 1: Learning experiment: Realized helmet profits and restocking rates

Panel A: Realized helmet profits



Panel B: Helmet restocking rates



Panel A reports realized helmet profits at endline among firms that adopted helmets within a month of the baseline survey, breaking down the sample by those that received access to returns versus not. Profit estimates are net of lost profits on items shops were unable to stock to afford helmets. Panel B reports rates of restocking among the same set of shops. The darker bar indicates that the shop purchased at least 1 additional stock of helmets by endline and reported an intent to stay in the market, and the lighter bar denotes shops that had yet to restock but reported planning to.

Table 1: Insurance experiment: Helmet stocking and contract uptake

Panel A: Effects of insurance offer on helmet stocking

	(1) Stocked (≤ 24H)	(2) Stocked	(3) Stocked (≤ 24H)	(4) Stocked	(5) Stocked (≤ 24H)	(6) Stocked	(7) Stocked (≤ 24H)	(8) Stocked
Offered insurance	0.075***	0.092**	0.073***	0.099**	0.002	-0.028	0.062*	0.112**
	(0.029)	(0.044)	(0.028)	(0.043)	(0.046)	(0.066)	(0.035)	(0.054)
High risk aversion					-0.066	-0.171***		
					(0.045)	(0.065)		
Offered insurance × High RA					0.168**	0.267***		
					(0.075)	(0.102)		
Observations	345	345	345	345	327	327	231	231
Control mean	0.047	0.203	0.047	0.203	0.047	0.203	0.033	0.167
Controls	Market	Market	Yes	Yes	Market	Market	Market	Market

Panel B: Insurance offer take-up and expected forgone returns among treated adopters

	Insurance uptake		Insurance expected payout − guaranteed		Insurance realized payout − guaranteed	
	(1) Stocked (≤ 24*H*)	(2) Stocked	(3) Dollars	(4) Share	(5) Dollars	(6) Share
𝔼[Sales]	-0.017	-0.036				
	(0.119)	(0.030)				
	{0.776}	{0.192}				
SD[Sales]	0.239	0.143**				
	(0.232)	(0.070)				
	{0.195}	{0.049}				
Accepted insurance			-0.024	-0.003	0.932	-0.044
			(0.285)	(0.063)	(2.765)	(0.268)
			{0.934}	{0.967}	{0.740}	{0.873}
Observations	22	51	51	51	47	47
Mean	0.500	0.392	-0.812	-0.113	-6.048	-0.493

$* p < 0.1$, $** p < .05$ $*** p < .01$. Robust standard errors in parentheses in Panel A. Bootstrapped standard errors in parentheses in Panel B (constructed with 10,000 draws). Bootstrapped p-values in braces.

This table reports results of the Insurance Experiment. In columns (1), (3), (5) and (7) the dependent variable equals 1 if the shop stocked helmets within 24 hours of the survey while columns (2), (4), (6) and (8) report effects on stocking after baseline, which could occur up to two weeks after the survey. Columns (7) - (8) restrict the sample to observations where the contract was also strictly dominated under the independent frequentist elicitation. Columns (3) and (4) include controls for industry, baseline revenue, days open per week, knowledge of a nearby helmet seller, and indicators for having space to store helmets, selling multiple products, and stocking a new product in the past year. High risk aversion indicates that the agent's coefficient of relative risk aversion, measured via a lottery choice game at the follow-up, exceeds the sample median. Panel B reports (endogenous) uptake of the insurance offer among treated enterprises that stocked helmets within 24 hours in column (1) and after baseline in (2). Columns (3) - (6) examine *insurance* shops that ever stocked helmets. Column (3) reports the expected value of the insurance offer less the guaranteed payment offered to firms, and column (4) reports column (3) normalized by the size of the guaranteed payment. Columns (5) and (6) are similar but use realized payouts rather than expectations. One market that was visited in both piloting and at baseline due to a sampling error is excluded from Panel A.

Table 2: Insurance experiment: Heterogeneity by firm size and age

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: Stocked within 24 hours			Dependent variable: Stocked		
	Employees	Firm profits	Firm age	Employees	Firm profits	Firm age
Treatment effect of insurance: smaller/younger firms						
Below median ×	0.025	0.020	0.036	0.046	0.039	0.043
Offered insurance	(0.032)	(0.045)	(0.034)	(0.047)	(0.060)	(0.060)
Treatment effect of insurance: larger/older firms						
Above median ×	0.197***	0.118**	0.120**	0.249**	0.173***	0.170**
Offered insurance	(0.071)	(0.047)	(0.054)	(0.100)	(0.066)	(0.066)
Pr(below = above)	0.039	0.160	0.219	0.070	0.130	0.169
Control mean (≤ median)	0.044	0.072	0.033	0.193	0.193	0.228
Control mean (> median)	0.054	0.027	0.062	0.243	0.216	0.175
Observations	344	316	345	344	316	345

$*\ p < 0.1$, $**\ p < .05$ $***\ p < .01$. Robust standard errors in parentheses.

This table reports heterogeneous treatment effects of *insurance* with respect to firm size and age. In columns (1) - (3) the dependent variable equals 1 if the shop stocked helmets within 24 hours of the survey, while in columns (4) - (6) the dependent variable captures stocking within the 2 week window provided after the baseline. The below median group of firms by employee size contains no paid employees, while the above median group includes all firms with at least 1. Median firm profits in the last month were PPP USD 325. The median enterprise age is 3 years. Pr(below=above) reports the p-value of the test that the treatment effects are equal. Control means capture the untreated average of the dependent variable within the respective groups. All estimates include market fixed effects and controls for industry, baseline revenue, days open per week, knowledge of a nearby helmet seller, and indicators for having space to store helmets, selling multiple products, and stocking a new product in the past year.

Table 3: Learning experiment: Treatment effects on stocking and entry

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Stocked in Phase 1		Ever Stocked		Stocked in Phase 2		Entrant	
Returns	0.096***	0.087***	0.085**	0.076**	0.070**	0.064**	0.069**	0.066**
	(0.029)	(0.028)	(0.034)	(0.032)	(0.031)	(0.030)	(0.028)	(0.027)
Supplier commitment	0.053*	0.060**	0.038	0.045	0.041	0.043	0.022	0.026
	(0.027)	(0.027)	(0.033)	(0.031)	(0.030)	(0.028)	(0.025)	(0.024)
Returns ×	-0.056	-0.041	-0.061	-0.042	-0.063	-0.046	-0.032	-0.021
Supplier commitment	(0.043)	(0.042)	(0.049)	(0.046)	(0.044)	(0.042)	(0.039)	(0.038)
Observations	929	929	929	929	929	929	929	929
Control mean	0.068	0.068	0.131	0.131	0.102	0.102	0.068	0.068
Controls	Strata FE	Yes	Strata FE	Yes	Strata FE	Yes	Strata FE	Yes

$* p < 0.1$, $** p < .05$, $*** p < .01$

This table reports treatment effects on helmet stocking from the Learning Experiment. Column (1)-(2) report effects on phase 1 stocking, supplied by the study or any other source. Columns (3)-(4) examine effects on stocking at any point prior to the endline survey. Columns (5)-(6) examine effects on phase 2 stocking. Column (7)-(8) examine effects on entry into the helmet market, defined by two or more stock purchases and reported intent to permanently stock helmets. The dependent variable in columns (5)-(8) is coded to 0 if a shop withdrew from the study because they did not wish to sell helmets or if the outcome was confirmed without the survey. All estimates include strata fixed effects. Columns where controls indicates "Yes" also include industry fixed effects, controls for distance to the manufacturer, log revenue, days open per week, and indicators for stocking a new product in the year before the baseline survey, selling multiple products at baseline, and having space to store helmets without stocking less of another item.

Table 4: Learning experiment: Relationship between beliefs and adoption

	SD Sales			log(1 + Var sales)		
	(1) LPM	(2) Logit	(3) LPM	(4) LPM	(5) Logit	(6) LPM
Returns	-0.032	0.290	0.232	-0.042	0.947	0.243
	(0.077)	(1.339)	(0.185)	(0.117)	(2.207)	(0.184)
E[sales]	0.021	0.479		0.069	1.906	
	(0.015)	(0.311)		(0.046)	(1.208)	
$\sigma(sales)$	-0.052**	-0.933**	-0.053**	-0.061**	-1.127**	-0.067**
	(0.023)	(0.406)	(0.024)	(0.027)	(0.506)	(0.030)
Returns × E[sales]	0.013	-0.145		0.038	-0.743	
	(0.034)	(0.394)		(0.110)	(1.556)	
Returns × $\sigma(sales)$	0.058	0.984*	0.096**	0.079	1.274*	0.121**
	(0.055)	(0.586)	(0.048)	(0.064)	(0.721)	(0.059)
Observations	304	271	304	304	271	304
Control mean	0.061	0.061	0.061	0.061	0.061	0.061
Controls	Full	Full	Full	Full	Full	Full
Expected Sale x Returns FEs	No	No	Yes	No	No	Yes

$*\ p < 0.1,\ **\ p < .05,\ ***p < .01$

Robust standard errors in parentheses. The dependent variable in all regressions is an indicator equal to 1 if the firm stocked helmets in the month after the baseline survey. In columns (1) - (3), $\sigma(sales)$ is the standard deviation of the agent's beliefs about helmet sales, and in columns (4) - (6) this value denotes the log of 1 plus the variance of sales. The sample excludes firms offered *supplier commitment*. The sample further excludes shops with 0 loss probability from stocking 5 helmets to avoid top censoring. Estimates include controls for storage space, education, baseline profits, respondent characteristics, and firm characteristics. Columns (3) and (6) control for bins of expected sales in 1 unit increments interacted with *returns*.

Table 5: Learning experiment: Instrumental variable estimates of stocking on posterior beliefs

	Instrumental Variables Estimates						Reduced form	
	(1) $\Delta\mathbb{E}[Sales]$	(2) $\Delta V[Sales]$	(3) $\Delta\mathbb{E}[Sales]$	(4) $\Delta V[Sales]$	(5) $\Delta\mathbb{E}[Sales]$	(6) $\Delta V[Sales]$	(7) $\Delta\mathbb{E}[Sales]$	(8) $\Delta V[Sales]$
Stocked (phase 1)	0.255	-1.552**	-0.018	-1.321*	0.121	-1.451**		
	(0.481)	(0.761)	(0.490)	(0.692)	(0.437)	(0.648)		
Returns							-0.003	-0.145*
							(0.059)	(0.082)
Supplier commitment							0.045	-0.051
							(0.054)	(0.075)
Returns × Supplier commitment							0.074	0.118
							(0.081)	(0.115)
Observations	820	820	832	832	1,652	1,652	1,652	1,652
Dep. var. mean	-0.142	0.075	-0.126	0.171	-0.134	0.124	-0.175	0.182
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F	3.40	3.40	5.32	5.32	4.51	4.51	–	–
Weak IV robust p-val	0.578	0.046	0.894	0.068	0.779	0.029		
Survey	ML	ML	EL	EL	Pooled	Pooled	Pooled	Pooled

$*\ p < 0.1,\ **\ p < .05,\ ***p < .01$

Robust standard errors in parentheses in columns (1) - (4). Standard errors clustered by firm in columns (5) - (8). The dependent variable mean in columns (7) - (8) is restricted to control firms. Weak instrument robust p-values are constructed using CLR inversion (Moreira, 2003). The dependent variables are the change in the inverse hyperbolic sine of the moment of beliefs of firm i since baseline, for instance $asinh(\mathbb{E}[\text{Sales at midline}]) - asinh(\mathbb{E}[\text{Sales at baseline}])$. Columns (1) - (6) instrument for stocking in the month following baseline using treatment assignment and treatment interacted with an indicator for a seller at baseline, controlling for the presence of a seller. The baseline seller interaction absorbs heterogeneity in the first stage, yielding better mean-squared error (Abadie et al., 2023). In columns (1) - (2), midline data is used, in columns (3) - (4) endline data is used, and in columns (5) - (8) both surveys are considered. All estimates include controls for the saturation of baseline helmet sellers in the market, firms' reported willingness to take risks to experiment, sample shops within a quarter km, the firm owner's performance on a digit span recall score, an indicator for whether the firm sold multiple products at baseline, and the firm's proximity to the helmet manufacturer. These controls were selected because they are likely to affect access to and retention of information affecting learning. Appendix Table A8 presents results with only strata and baseline seller controls. Using CHS post-lasso-orthogonalized controls, selected from a list including those selected in the main specification as well as a wider list, yields a coefficient on endline expectations of $.015 (p = .975)$ and one on endline variance of $-1.16 (p = .090)$.

Table 6: Information treatment: Effects on helmet uptake

		Helmet sales		Helmet revenue	
	(1)	(2)	(3)	(4)	(5)
1(Information treatment)	0.022**	0.030**	0.028**	0.029*	0.039**
	(0.009)	(0.015)	(0.014)	(0.017)	(0.017)
Signal > Expectation		-0.014		-0.011	
		(0.021)		(0.023)	
Signal > Median			-0.011		-0.030
			(0.017)		(0.018)
Expectation		-0.000		0.001	
		(0.000)		(0.003)	
Observations	727	722	727	722	727
Control mean	0.006	0.006	0.006	0.006	0.006

$* p < 0.1$, $** p < .05$, $*** p < .01$

Robust standard errors in parentheses. The dependent variable is an indicator for ordering helmets at the endline survey and the sample is restricted to shops that had not stocked helmets prior to that point. 1(Information treatment) indicates that a shop received helmet sale and price data from 5 randomly selected shops. Signal > expectation denotes that the signal average sales or revenue value exceeded the respondent's beliefs and is 0 otherwise or if the shop received no information, and Signal > median is one if the average sales or revenue value was greater than the median value across signals and 0 otherwise or if the shop received no signal. Columns (2) - (3) consider signals about helmet sales, and columns (4) - (5) examine signals about helmet revenue. All estimates include controls for knowing of a helmet seller at midline, revenue in the month before the survey, and an indicator equal to 1 if the shop was affected by floods that occurred near midline.

Table 7: Spillover survey: Effect of helmet entrant on neighbor adoption

	(1)	(2)	(3)	(4)
	Know seller in market		Ordered helmets	
Panel A: Motorcycle shops				
Baseline (treated) market	0.115*	0.133**	0.098***	0.104***
	(0.069)	(0.061)	(0.035)	(0.036)
Observations	376	376	376	376
Control mean	0.361	0.361	0.091	0.091
Controls	No	Yes	No	Yes
Panel B: All shops				
Baseline (treated) market	0.122**	0.143***	0.048**	0.059***
	(0.052)	(0.049)	(0.022)	(0.022)
Observations	665	665	665	665
Control mean	0.300	0.300	0.070	0.070
Controls	No	Yes	No	Yes

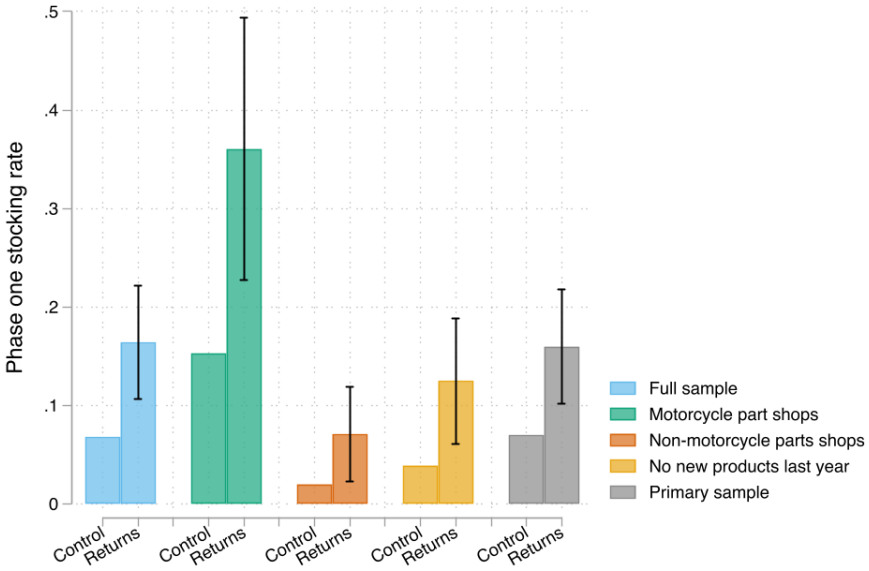
$*\ p < 0.1,\ **\ p < .05,\ ***p < .01$

Standard errors in parentheses clustered by market. Panel A restricts the sample to shops selling motorcycle-related products and Panel B includes all firms. The variable "Baseline (treated) market" equals 1 if the market was randomly selected for surveys at baseline, and is 0 if the market was a pure control and was skipped. All results are from the spillover survey with non-baseline shops, collected 3-4 months after baseline. The dependent variable in columns (1)-(2) equals 1 if the shop reported knowing a helmet seller in their market (selling the study's variety, or any other variety). The dependent variable in columns (3)-(4) equals 1 if the shop purchased helmet stock. All estimates include county fixed effects. Estimates with controls include additional covariates for revenue, business age, employees, and the number of products the firm sells. All estimates exclude 35 observations from shops that started selling helmets before the study began. These shops were ineligible but initially misreported their sales start date to obtain subsidized stock. Back checks were conducted after delivery (when no misreporting incentive is present) with all shops selling helmets at the time of the survey to confirm eligibility, which is the full set in which prior sales are detectable, since a shop selling helmets before the study would still have been selling them. Shops that revealed they were ineligible during these back checks are excluded. Exclusions are more common in pure control markets, since surveyors who had visited baseline markets already knew which shops sold helmets and did not recruit them. Any undetected ineligible sellers are therefore concentrated in pure control markets, which biases against detecting a spillover.

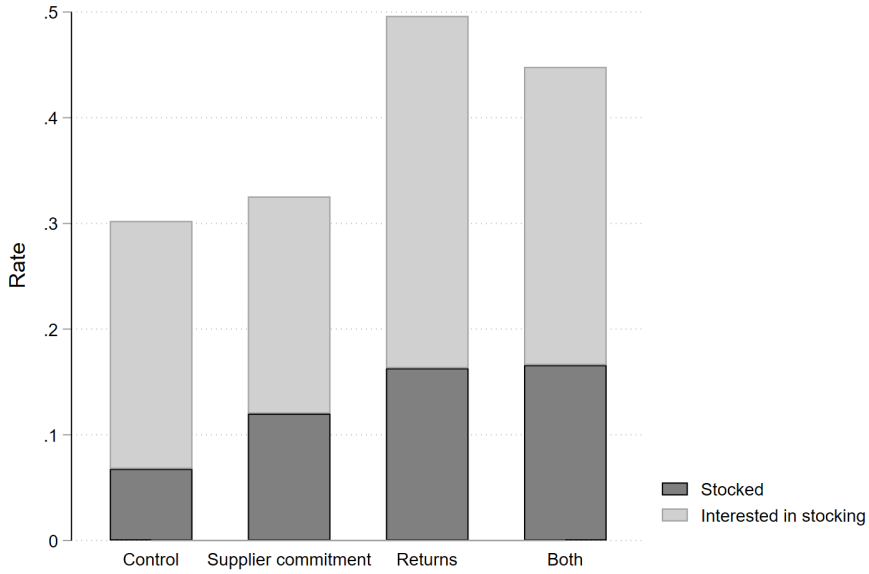
Appendix

Figure A1: Learning experiment: Treatment effects on phase one helmet adoption plus interest

Panel A: Returns effects in sub-samples

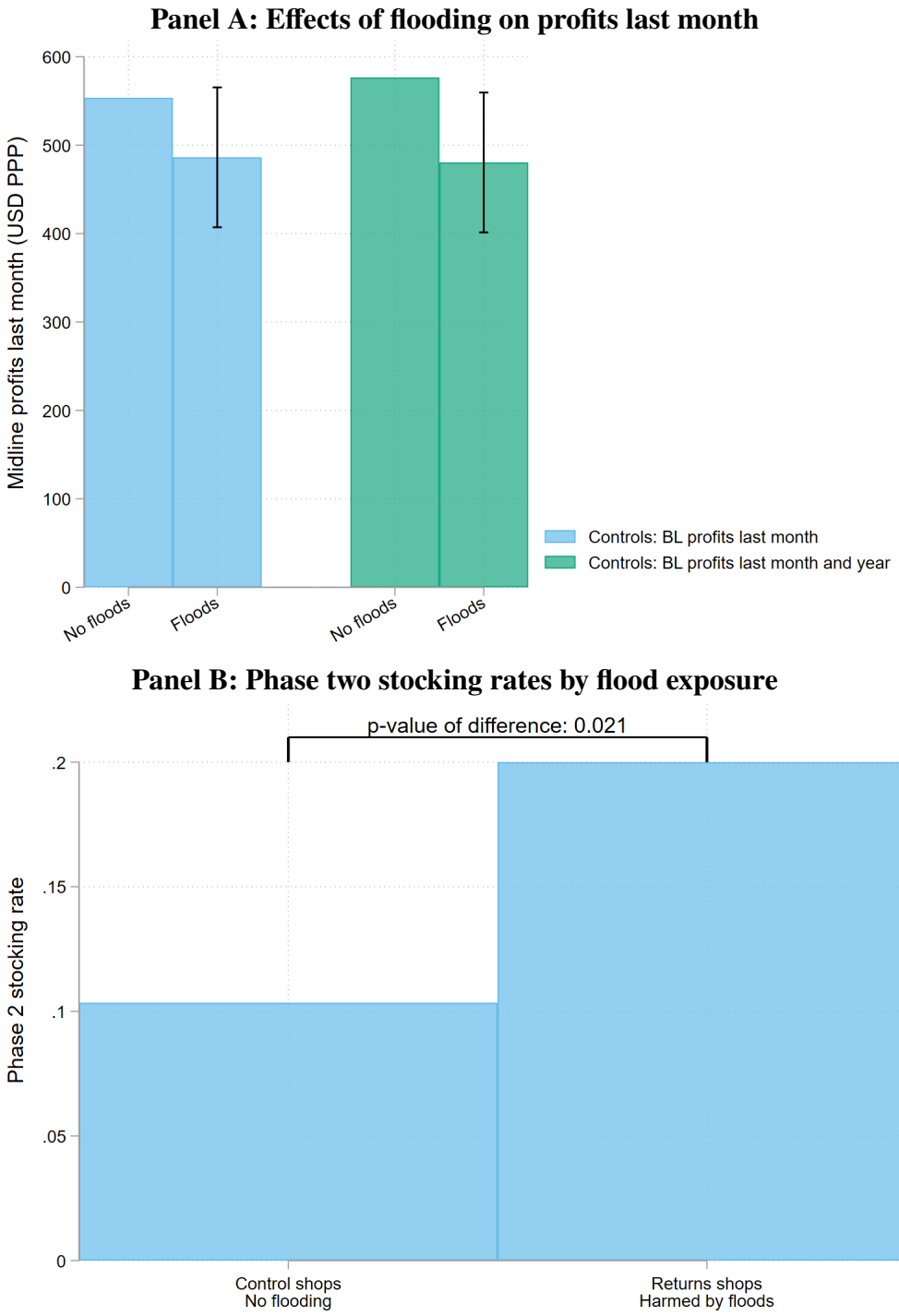


Panel B: Phase one stocking plus interest



Panel A plots *returns* effects on phase one stocking in five groups: the full sample (N = 929), motorcycle spare parts shops (N = 337), non-motorcycle parts shops (N = 592), shops that did not experiment with stocking a new product the year before the survey (N = 614), and shops in the primary (versus replacement) sampling pool (N = 900). The first bar is the control mean, the second bar is the control mean plus *returns* effect where the bars denote 95% confidence intervals, estimated from a regression of phase one stocking on *returns*, the *supplier commitment*, an interaction between the two, and strata fixed effects. Panel B reports the rate of shops that stocked helmets in phase one, from the study or a different source, or that requested the field officer call them back in two days for a final purchase decision. The first bar presents the control stocking rate. The second reports the adoption rate among shops receiving the *supplier commitment*. The third bar reports adoption with *returns*. The fourth bar reports the adoption rate conditional on receiving both treatments.

Figure A2: Learning experiment: Phase two stocking effects by flood exposure



Panel A plots profits the month before the midline survey among shops that reported they were negatively impacted by severe flooding and those that were not. Both results control for strata and industry fixed effects, controls for distance to the manufacturer, log revenue, days open per week, and indicators for stocking a new product in the year before the baseline survey, selling multiple products at baseline, and having space to store helmets without stocking less of another item. The first estimate further controls for baseline profits last month, and the second controls for baseline monthly and annual profits. Error bars on the flood estimates show 95% confidence intervals. Panel B reports phase two stocking rates among control shops not affected by flooding and returns shops affected by flooding. The p-value for equality is from a regression of stocking on returns among these firm types, including the controls used in Panel A (excluding baseline profits due to missing values).

Table A1: Summary statistics and baseline balance

	Insurance experiment		Learning experiment Returns		Learning experiment Supplier commitment	
Variable	(1) Control mean [SD]	(2) T - C	(3) Control mean [SD]	(4) T - C	(5) Control mean [SD]	(6) T - C
Respondent age	34.18 [10.86]	0.77 (1.12)	32.50 [9.38]	0.77 (0.62)	33.30 [9.21]	-0.75 (0.62)
Female	0.30 [0.46]	0.03 (0.05)	0.28 [0.45]	-0.03 (0.03)	0.27 [0.44]	0.00 (0.03)
Years of education	12.78 [2.79]	0.19 (0.27)	13.23 [2.48]	0.16 (0.16)	13.34 [2.54]	-0.02 (0.17)
Business age	4.53 [5.46]	-0.22 (0.52)	5.29 [5.45]	0.18 (0.36)	5.68 [5.77]	-0.57 (0.36)
Motorcycle spares shop	0.59 [0.49]	-0.01 (0.05)	0.37 [0.48]	-0.02 (0.03)	0.36 [0.48]	-0.00 (0.03)
Revenue last month	1,235.46 [1,482.82]	-132.01 (152.00)	1,459.28 [2,107.67]	153.17 (159.08)	1,590.50 [2,458.95]	-125.91 (168.67)
Costs last month	972.27 [1,280.78]	15.70 (185.11)	655.92 [835.61]	138.05* (79.85)	747.21 [1,045.04]	-49.55 (83.74)
Profits last month	508.77 [593.92]	-46.39 (57.29)	622.90 [977.04]	61.17 (66.44)	674.88 [937.10]	-32.39 (66.60)
1(paid employees)	0.22 [0.41]	0.05 (0.04)	0.25 [0.43]	-0.01 (0.03)	0.25 [0.43]	-0.01 (0.03)
Wage bill last week	58.26 [348.89]	10.85 (29.99)	34.69 [164.33]	-5.05 (10.47)	21.60 [47.16]	18.33** (9.01)
Hours open/week	77.41 [19.50]	0.24 (1.57)	85.35 [15.37]	-0.16 (0.95)	86.06 [15.07]	-1.74* (0.95)
KM to helmet seller	NA –	NA –	2.24 [3.17]	0.06 (0.15)	2.28 [3.27]	-0.06 (0.15)
Know helmet seller	0.56 [0.50]	-0.10** (0.05)	0.25 [0.44]	0.03 (0.03)	0.27 [0.44]	0.00 (0.03)
Min. to motorcycle taxi stand	NA –	NA –	3.16 [6.01]	-0.07 (0.36)	3.00 [5.77]	0.24 (0.36)
New product in last year	0.28 [0.45]	0.03 (0.04)	0.34 [0.48]	-0.01 (0.03)	0.36 [0.48]	-0.05 (0.03)
E[sales]	3.43 [1.73]	0.03 (0.15)	3.75 [2.01]	0.06 (0.13)	3.89 [2.06]	-0.21 (0.14)
V(sales)	2.96 [2.88]	-0.13 (0.25)	1.26 [1.84]	0.08 (0.12)	1.33 [1.80]	-0.06 (0.12)
Observations	172	173	461	468	463	466
Joint p-value		0.49		0.89		0.46

$*\ p < 0.1$, $**\ p < .05$ $***\ p < .01$. Standard deviations in brackets. Standard errors in parentheses.

Column (1) reports the mean and standard deviation of the indicated variable across control enterprises in the Insurance Experiment. Columns (3) and (5) report the same value across control surveys in the *returns* and *supplier commitment* arms of the Learning Experiment. Columns (2), (4) and (6) report the difference between the treatment and control group in the sample and arm as denoted in the table header, estimated via OLS including strata fixed effects. The last row reports the p-value associated with a test for joint orthogonality, constructed by estimating a seemingly unrelated regression model then estimating a Wald test to allow for missing variables.

Table A2: Summary statistics and balance
Spillover survey

	Motorcycle shops		All shops	
	(1)	(2)	(3)	(4)
Variable	Control mean [SD]	T - C	Control mean [SD]	T - C
Respondent age	35.04	0.90	35.20	-0.00
	[10.41]	(0.99)	[10.59]	(0.82)
Business age	4.58	-0.33	4.95	-0.45
	[4.11]	(0.48)	[5.04]	(0.39)
Revenue last month	990.57	14.04	982.30	-60.27
	[759.23]	(101.57)	[805.51]	(82.68)
Costs last month	729.83	-53.75	710.49	-24.51
	[825.97]	(91.45)	[812.68]	(81.38)
Profits last month	465.55	-8.88	458.62	-27.24
	[342.76]	(50.30)	[349.98]	(41.93)
1(Employees)	0.30	-0.04	0.29	-0.03
	[0.46]	(0.06)	[0.45]	(0.04)
Helmet storage capacity	22.63	-10.77*	21.46	-2.12
	[57.87]	(6.07)	[51.71]	(6.64)
Owner hours of work/week	76.55	-1.30	76.71	-1.64
	[15.39]	(1.98)	[15.68]	(1.83)
Motorcycle spares shop	NA	NA	0.67	-0.20***
	–	–	[0.47]	(0.05)
Observations	219	157	327	338
Joint p-value		0.51		0.01

$*\ p < 0.1$, $**\ p < .05$ $***\ p < .01$. Standard deviations in brackets. Standard errors in parentheses clustered by market.

Column (1) reports the mean and standard deviation of the indicated variable across enterprises in pure control markets, restricting the sample to motorcycle spare part/repair shops. Column (3) is similar but with no sample restriction. Columns (2) and (4) report the difference between the treatment and control group in the sample and arm as denoted in the table header, estimated via OLS including county fixed effects. The last row reports the p-value associated with a test for joint orthogonality, constructed by estimating a seemingly unrelated regression model then estimating a Wald test to allow for missing variables. 35 shops that sold motorcycle helmets prior to the beginning of the study are excluded.

Table A3: Baseline beliefs about helmet profitability

	Mean	SD	25th percentile	50th percentile	75th percentile	Obs
Panel A: Insurance experiment						
Pr(Loss| Stock 10 helmets)	0.307	0.233	0.100	0.300	0.500	350
Pr(Helmets profits > current stock)	0.504	0.219	0.400	0.500	0.600	349
Pr(Sell out in 8 weeks|Stock)	0.442	0.256	0.200	0.450	0.600	350
$\mathbb{E}$[8 week revenue] − stock cost	0.373	15.986	-8.850	2.294	10.672	340
8 week expected sales	3.466	1.825	2.075	3.250	4.600	340
8 week SD sales	1.489	0.824	0.829	1.303	2.041	340
Panel B: Learning experiment						
Pr(Loss | Buy 10 helmets)	0.369	0.241	0.200	0.300	0.500	922
Pr(Helmets profits > current stock)	0.506	0.214	0.400	0.500	0.600	922
Pr(Representative firm restocked)	0.437	0.198	0.333	0.444	0.556	920
$\mathbb{E}$[1 month revenue] − stock cost	-5.078	29.082	-28.330	-3.384	16.139	873
1 month expected sales	3.765	2.052	2.300	3.500	5.000	873
1 month SD sales	0.928	0.658	0.500	0.829	1.221	873

This table reports baseline beliefs about helmet profitability. The first row in each panel is the firm's belief about the likelihood that they would lose money, inclusive of the opportunity cost of funds, if they stocked 10 helmets. The second row is the likelihood that stocking 10 helmets would raise the firm's profits, inclusive of any losses from stocking less of other items. The third row in Panel A denotes the firm's belief about their probability of selling 3 helmets in 8 weeks if they stocked them, and in Panel B the row denotes the enterprise's perceived likelihood that a representative shop would restock if given helmets to learn about the market. The fourth row captures expected revenue net of stocking cost for the study offers over an 8 week period in Panel A and a 1 month period in Panel B. The final two rows present beliefs about expected sales and the standard deviation of sales, measured with a frequentist mechanism, over an 8 week and 1 month period respectively.

Table A4: Insurance experiment: CRRA heterogeneity robustness

	(1) CRRA	(2) $\mathbb{I}$(CRRA > P50)	(3) Stocked	(4) Stocked	(5) Stocked ($\leq$ 24H)	(6) Stocked ($\leq$ 24H)
Offered insurance	-0.119	-0.049	0.025	0.015	0.022	0.018
	(0.222)	(0.059)	(0.064)	(0.067)	(0.045)	(0.050)
$\mathbb{I}$(CRRA $\geq$ P75)			-0.106		-0.071	
			(0.065)		(0.045)	
Offered insurance $\times$			0.184*		0.144**	
$\mathbb{I}$(CRRA $\geq$ P75)			(0.102)		(0.072)	
CRRA				-0.034**		-0.019*
				(0.016)		(0.011)
Offered insurance $\times$				0.049*		0.036*
CRRA				(0.027)		(0.020)
Observations	327	327	327	327	327	327
Control Mean	1.726	0.476	0.203	0.203	0.047	0.047
Within R^2	0.001	0.003	0.030	0.033	0.040	0.039

Robust standard errors in parentheses. $*\ p < 0.1$, $**\ p < .05$, $***p < .01$

This table reports results of the Insurance Experiment. Column (1) reports the effect of *insurance* on the CRRA parameter estimated from the lottery choice game. Column (2) is similar but the dependent variable is an indicator for above median risk aversion on the game, used in heterogeneity analysis in Table 1, instead of the continuous value. The dependent variable in columns (3) - (4) captures ever stocking, and (5) - (6) examine stocking within 24 hours. Columns (3) and (5) look at heterogeneity with an indicator for the CRRA measurement being at least the 75th percentile, and (4) and (6) look at heterogeneity with the continuous measure. All estimates include market fixed effects and exclude one market that was visited in piloting due to a sampling error.

Table A5: Learning experiment: Heterogeneous effects by presence of a pre-existing seller

	(1)	(2)	(3)	(4)
	Stocked in Phase 1		Stocked in Phase 2	
Returns	0.127***	0.126***	0.078**	0.077**
	(0.032)	(0.031)	(0.034)	(0.033)
Supplier commitment	0.085***	0.085***	0.069**	0.055*
	(0.029)	(0.028)	(0.032)	(0.030)
Returns × Supplier commitment	-0.099**	-0.080*	-0.080	-0.052
	(0.048)	(0.046)	(0.049)	(0.047)
BL seller	0.125**	0.096**	0.092*	0.033
	(0.050)	(0.047)	(0.053)	(0.048)
Returns × BL seller	-0.122	-0.149**	-0.038	-0.052
	(0.075)	(0.072)	(0.080)	(0.074)
Supplier commitment × BL seller	-0.128*	-0.103	-0.112	-0.046
	(0.073)	(0.071)	(0.077)	(0.071)
Returns × Commitment	0.166	0.148	0.074	0.024
× BL seller	(0.107)	(0.104)	(0.111)	(0.106)
Observations	929	929	929	929
Control mean (no BL seller)	0.034	0.034	0.073	0.073
Controls	Strata FE	Yes	Strata	Yes

$*\ p < 0.1,\ **\ p < .05,\ ***p < .01$

This table reports heterogeneous treatment effects on helmet stocking from the Learning Experiment based on proximity to an existing helmet seller. "BL Seller" indicates that a shop reported that a pre-existing helmet seller operated near them at baseline. Columns (1)-(2) reports effects on phase 1 stocking, supplied by the study or any other source. Column (3)-(4) report phase 2 stocking effects. The dependent variable in columns (3)-(4) is coded to 0 if a shop withdrew from the study because they did not wish to sell helmets or if the outcome was confirmed without the survey. All estimates include strata fixed effects. Columns (2) and (4) include industry fixed effects, controls for distance to the manufacturer, log revenue, days open per week, and indicators for stocking a new product in the year before the baseline survey, selling multiple products at baseline, and having space to store helmets without stocking less of another item.

Table A6: Learning experiment: Belief-adoption relationship robustness to controls and sample

	Sample: Pr(Loss|Stock 5) > 0			All	Non-degenerate	All
	(1) Revenue	(2) Sales	(3) Sales	(4) Sales	(5) Sales	(6) Sales
Returns	0.032	-0.006	0.024	0.015	0.022	-0.022
	(0.201)	(0.073)	(0.072)	(0.067)	(0.086)	(0.076)
$\mathbb{E}$[sales/revenue]	0.009	0.021	0.021	0.008	0.023*	0.023
	(0.007)	(0.015)	(0.015)	(0.008)	(0.013)	(0.015)
σ(sales/revenue)	-0.028**	-0.042**	-0.031*	-0.022	-0.054**	-0.054**
	(0.012)	(0.020)	(0.018)	(0.017)	(0.024)	(0.022)
Returns ×	0.003	0.014	0.008	0.013	-0.018	0.013
$\mathbb{E}$[sales/revenue]	(0.016)	(0.033)	(0.033)	(0.015)	(0.021)	(0.033)
Returns ×	0.029	0.039	0.032	0.023	0.092*	0.051
σ(sales/revenue)	(0.029)	(0.053)	(0.053)	(0.039)	(0.053)	(0.054)
Pr(Loss|Stock 5) = 0						-0.071
						(0.109)
Returns ×						0.239
Pr(Loss|Stock 5) = 0						(0.220)
$\mathbb{E}$[sales/revenue] ×						-0.010
Pr(Loss|Stock 5) = 0						(0.024)
σ(sales/revenue)						0.101
× Pr(Loss|Stock 5) = 0						(0.086)
Returns × $\mathbb{E}$[sales/revenue]						-0.022
× Pr(Loss|Stock 5) = 0						(0.047)
Returns × σ(sales/revenue)						-0.136
× Pr(Loss|Stock 5) = 0						(0.146)
Observations	304	304	304	427	286	427
Control mean	0.061	0.061	0.061	0.068	0.069	0.068
Controls	Full	DPLASSO	None	Full	Full	Full

$* p < 0.1$, $** p < .05$ $*** p < .01$

Robust standard errors in parentheses. This table reports the relationship between helmet stocking and belief uncertainty under different measures and controls. The dependent variable equals 1 if the firm stocked helmets during phase 1. The sample excludes *supplier commitment* firms. In column (1), the independent variables are expected helmet net revenue divided by 10 (for display) and the standard deviation of this value. Columns (2)-(6) examine expected sales and SD sales. Columns (1)-(3) exclude shops with 0 loss probability from stocking 5 helmets. Columns (4)-(6) include all observations, but column (5) screens for comprehension of belief elicitation by dropping respondents reporting degenerate belief distributions (Var(sales) ≤ 0.25). Covariates in column (2) were selected with double post LASSO. Columns (1) and (4)-(6) include controls for storage space, education, baseline profits, respondent characteristics, and firm characteristics.

Table A7: Learning experiment: Beliefs of adopters with versus without returns

	(1) Belief levels	(2) Belief logs
$\mathbb{E}$[sales/revenue]	-0.001	
	(0.036)	
	{0.987}	
σ(sales/revenue)	0.202*	
	(0.108)	
	{0.060}	
log(1 + E[sales])		-0.012
		(0.186)
		{0.943}
log(1 + V(sales))		0.246**
		(0.121)
		{0.044}
Know helmet seller	-0.309**	-0.313**
	(0.148)	(0.147)
	{0.036}	{0.034}
Observations	46	46

$*\ p < 0.1,\ **\ p < .05\ ***\ p < .01$

Bootstrapped standard errors in parentheses (10,000 draws). Bootstrapped p-values in braces. This table reports beliefs about demand for helmets of adopters with and without *returns*. The dependent variable in each column is an indicator equal to 1 if the agent received *returns*, and the sample is restricted to those that stocked helmets in phase one and did not receive the supplier commitment offer.

Table A8: Learning experiment: Learning robustness

Panel A: Effects of stocking on posterior beliefs, robustness

	Instrumental Variables Estimates						OLS	
	(1) $\Delta asinh$ $\mathbb{E}[Sales]$	(2) $\Delta asinh$ $V[Sales]$	(3) $\Delta\mathbb{E}[Sales]$	(4) $\Delta SD[Sales]$	(5) $\Delta\mathbb{E}[Net\ Rev]$	(6) $\Delta SD[Rev]$	(7) $\Delta asinh$ $\mathbb{E}[Sales]$	(8) $\Delta asinh$ $V[Sales]$
Stocked (phase 1)	0.145	-1.234*	0.395	-1.206**	8.425	-25.942**	-0.021	-0.189**
	(0.430)	(0.633)	(1.593)	(0.591)	(33.431)	(12.357)	(0.068)	(0.093)
Observations	1,652	1,652	1,652	1,652	1,652	1,652	1,652	1,652
Dep. var. mean	-0.134	0.124	-0.459	0.109	-7.586	2.671	-0.134	0.124
Controls	Strata FE	Strata FE	Yes	Yes	Yes	Yes	Strata FE	Strata FE
Kleibergen-Paap F	4.81	4.81	4.51	4.51	4.51	4.51	–	–
Weak IV robust p-val	0.744	0.059	0.896	0.054	0.912	0.049	NA	NA

Panel B: Updating correlations with entry (phase one adopters). Dependent variable: endline entrant

	Independent variables: Beliefs about sales				Beliefs about revenue			
$\Delta\mathbb{E}$[Sales/Revenue]	0.023	0.024	0.015	0.017	0.012	0.012	0.006	0.004
	(0.021)	(0.021)	(0.023)	(0.023)	(0.011)	(0.011)	(0.011)	(0.011)
ΔSD[Sales/Revenue]	-0.113**	-0.086*	-0.132**	-0.106	-0.058**	-0.046*	-0.072**	-0.074**
	(0.047)	(0.050)	(0.060)	(0.064)	(0.024)	(0.025)	(0.029)	(0.031)
Observations	102	102	102	102	102	102	102	100
Strata FE	No	No	Yes	Yes	No	No	Yes	Yes
Controls for realization	No	Yes	No	Yes	No	Yes	No	Yes

$*\ p < 0.1,\ **\ p < .05,\ ***p < .01$

Standard errors clustered by firm in parentheses in Panel A, robust in Panel B. Panel A reports relationships between phase one stocking and belief updating. Weak IV robust p-values use CLR inversion (Moreira, 2003). The dependent variable in columns (1) - (2) and (7) - (8) is the change in $asinh$ of the moment of beliefs of firm i since baseline. In columns (3) - (6) the dependent variable is the change in a firm's expectation or standard deviation in beliefs since baseline. Columns (1) - (6) instrument for phase one stocking using treatment assignment and treatment interacted with an indicator for a seller at baseline, controlling for the presence of a seller. Columns (7) - (8) report OLS estimates of the change in beliefs on phase 1 stocking. I pool data from the midline and endline surveys. Estimates where controls is "yes" control for the baseline saturation of nearby helmet sellers, reported willingness to take risks to experiment, digit span recall, an indicator for whether the firm sold multiple products at baseline, and the firm's proximity to the helmet manufacturer, in addition to strata fixed effects. Controls likely to affect learning were selected. Panel B reports relationships between endline entry (stocking at least twice plus intent to stay in the market) and the change in firms' beliefs from baseline to midline among phase one adopters. I control for 2 helmet bins of expected midline sales. Even columns control for realized helmet sales/revenue. The dependent variable mean is 0.632.

Table A9: Learning experiment: Helmet sales outcomes vs expectations (phase 1 adopters)

Panel A: Levels

	Sales		Revenue	
	(1)	(2)	(3)	(4)
	BL → ML	ML → EL	BL → ML	ML → EL
Expectation at BL	0.169		0.199	
	(0.123)		(0.159)	
Expectation at ML		0.544***		0.463**
		(0.119)		(0.217)
Constant	3.229***	1.199***	67.270***	29.601**
	(0.603)	(0.431)	(16.052)	(14.219)
Observations	107	111	109	116
Adjusted R^2	0.011	0.179	0.007	0.073
Pr(η BL= η ML)		0.032		0.300

Panel B: Elasticities

	log(Sales)		log(Revenue)	
	(1)	(2)	(3)	(4)
	BL → ML	ML → EL	BL → ML	ML → EL
log(Expectation at BL)	0.279**		0.239*	
	(0.122)		(0.131)	
log(Expectation at ML)		0.690***		0.364***
		(0.139)		(0.114)
Constant	0.921***	0.425***	3.298***	2.703***
	(0.184)	(0.139)	(0.587)	(0.463)
Observations	98	91	97	85
Adjusted R^2	0.045	0.239	0.028	0.099
Pr(η BL= η ML)		0.047		0.473

$*\ p < 0.1,\ **\ p < .05\ ***\ p < .01$

Robust standard errors in parentheses. This table reports regressions of sales outcomes and revenue on expectations elicited from respondents before the indicated period. Panel A reports estimates in levels and Panel B in terms of elasticities. All estimates include only observations where the shop stocked helmets in phase 1. Columns (1)-(2) report beliefs about sales while columns (3)-(4) consider beliefs about revenue, constructed from expected sales and expected prices. Beliefs were elicited using a frequentist method. Columns (1) and (3) report regressions of sales/revenue from baseline to midline on baseline expectations. Columns (2) and (4) report regressions of midline to endline realizations on midline expectations. The p-values in columns (2) and (4) report results of a test that the baseline and midline elasticities are equal, estimated using seemingly unrelated regressions.

Table A10: Learning experiment: Helmet profit dynamics and selling costs

	(1) Helmet profits per month EL	(2) Total costs selling helmets	(3) Helmet costs net est. stock	(4) Any fixed costs	(5) Fixed costs	(6) Δ capacity utilization
Months since first stocked	-15.561					
	(9.390)					
Estimated cost of helmet stock (PPP)		0.453***				
		(0.169)				
Returns			-7.928	0.050	-0.632	
			(8.486)	(0.060)	(2.009)	
Stocked in phase 1						0.046*
						(0.024)
Observations	96	132	132	128	128	712
Dep. var. mean	28.187	73.735	-10.416	0.133	2.381	0.064

$*\ p < 0.1,\ **\ p < .05\ ***\ p < .01$

Robust standard errors in parentheses. The sample in column (1) consists of shops that ordered helmets by midline and reported that they planned to stay in the market by endline. The sample in columns (2) - (5) consists of shops that reported selling helmets. Column (1) regresses the difference in reported helmet profits from endline versus midline on the shop's tenure in the helmet market at endline, controlling for the number of days between the midline and endline surveys. Column (2) regresses total reported costs of stocking helmets on the cost of the helmet stock, measured via administrative data or estimated. The dependent variable in column (3) equals reported helmet costs net of stock price. Column (4) examines whether shops reported any fixed costs of helmet sales, and column (5) reports total fixed costs. Column (6) reports the change in capacity utilization of the firm from baseline to endline, measured as worst week over best week profits in the last month.

Table A11: Learning experiment: Helmet adoption rates and proximity of other sellers

	(1) Ordered Baseline	(2) Ordered Midline	(3) Ordered Endline	(4) Ordered	(5) Observed Seller
Know helmet seller	0.129**				0.325***
	(0.055)				(0.034)
Know of large seller, BL	-0.198***				
	(0.068)				
Noticed seller by ML		0.179**			
		(0.085)			
Noticed helmet seller by EL			0.201***		
			(0.063)		
Study adopter within 0.25km				0.031**	0.138***
				(0.014)	(0.031)
Sample shops in quarter km				-0.001***	
				(0.000)	
Study adopter 0.25-0.5km					-0.003
					(0.030)
Observations	231	165	165	771	929
Control mean	0.034	0.028	0.029	0.010	0.237
Controls	DPLASSO	DPLASSO	DPLASSO	DPLASSO	None
Sample restriction	Control	No BL order control	No BL order control	No BL order	

$* p < 0.1$, $** p < .05$, $*** p < .01$

Robust standard errors in parentheses. This table reports correlations between proximity to a shop selling helmets and the firm's own stocking decisions. In column (1), the dependent variable equals 1 if the shop stocked in phase one, in column (2) the dependent variable captures stocking between a month after baseline and a month after midline, column (3) looks at stocking more than a month after midline, and column (4) captures ever stocking. Column (5) indicates that the shop reported knowing a helmet seller in their location at midline. Noticed seller indicates that the shop did not know of a shop selling helmets before the indicated survey, then observed one. Columns (1) - (3) include only untreated shops, with (2) and (3) further excluding those that stocked in the 4 weeks after baseline. Column (4) includes all firms that did not stock helmets within a month of baseline. Column (1)-(4) includes enterprise and shopkeeper demographic controls selected using double-post selection LASSO. Control means in columns (1)-(4) are conditional on not knowing of a seller.

Table A12: Learning experiment: Effect of returns on helmet access

	(1)	(2)	(3)	(4)	(5)	(6)
	Helmet shop in market at endline	Ever helmet shop in market	Cumulative helmets stocked at endline		Cumulative helmets sold at endline	
Returns	0.078**	0.133***	1.822**	2.789***	1.354*	2.103**
	(0.038)	(0.041)	(0.833)	(1.012)	(0.746)	(0.950)
Supplier commitment	0.043	0.083**	0.647	1.286***	0.311	0.766**
	(0.038)	(0.042)	(0.481)	(0.460)	(0.349)	(0.343)
Returns × Commitment	-0.035	-0.095	-1.413	-2.715**	-1.224	-2.104**
	(0.054)	(0.058)	(0.976)	(1.155)	(0.827)	(1.041)
BL Seller				2.099		1.545
				(1.320)		(0.966)
Returns × BL Seller				-3.698**		-2.880**
				(1.747)		(1.357)
Commitment × BL Seller				-2.591		-1.873
				(1.578)		(1.141)
Returns × Commitment				4.928**		3.382**
× BL seller				(2.356)		(1.689)
Observations	929	929	929	929	926	926
Control mean	0.242	0.314	1.525	0.689	0.966	0.407

$*\ p < 0.1$, $**\ p < .05$, $***p < .01$

Robust standard errors in parentheses. This table reports evidence of the effect of the intervention on helmet access. Column (1) equals 1 if the respondent shop entered the helmet market by endline or reported that a shop near them sold helmets. Column (2) is similar but equals 1 if the respondent ever stocked helmets or ever reported that a shop near them stocked helmets. Columns (3) - (4) examine the total number of helmets stocked by the shop by endline, and columns (5) - (6) examine total helmet sales by endline. In 3 cases, shops stocked helmets and did not complete the endline survey because their enterprises closed, so endline sales are imputed using midline values. Estimates include strata fixed effects and controls for industry, proximity to the manufacturer, days open per week at baseline, log baseline revenue, indicators for adopting a new product in the year before the survey, selling multiple products at baseline and having space to store helmets. I also control for exposure to floods at midline.

A Model Details

A.1 Lagrangian and solution to optimal investment

The Lagrangian of the entrepreneur's optimization problem may be written as

$$\begin{aligned}\mathcal{L} = u(c_t) + \lambda_t \left[y_t - c_t - a_t - w_s I_{st} - w_n I_{nt}\right] + \kappa_{at}\left[a_t - \underline{a}\right] + \kappa_{\chi t}\left[I_{nt}\cdot(I_{nt}-\chi)\right] \\ + \iota_{st} I_{st} + \iota_{nt} I_{nt} + \delta\,\mathbb{E}_t\left[V^*(y_{t+1},\Gamma,\theta) - \bar{R}(y_{t+1},\mathcal{I}_{t+1}(I_{nt}),\Gamma)\middle|\mathcal{I}_t\right]\end{aligned} \tag{14}$$

Differentiating, we get the set of first order conditions

$$\begin{aligned}&\mathcal{L}_c : u'(c_t) = \lambda_t \\ &\mathcal{L}_a : \lambda_t = \mathbb{E}_t\,\lambda_{t+1} + \kappa_{at} \\ &\mathcal{L}_{I_s} : \lambda_t w_s - \iota_{st} = \delta\,\mathbb{E}_t\left[\lambda_{t+1}\frac{\partial}{\partial I_{st}}\pi_s(I_{st},\nu_{st+1})\right] \\ &\mathcal{L}_{I_n} : \lambda_t w_n - \kappa_{\chi t}(2I_{nt}-\chi) - \iota_{nt} = \delta\,\mathbb{E}_t\left[\lambda_{t+1}\frac{\partial}{\partial I_{nt}}\pi_n(I_{nt},\nu_{nt+1}+\theta)\right] - \delta\frac{\partial}{\partial I_{nt}}\mathbb{E}_t\left[\bar{R}(y_{t+1},\mathcal{I}_{t+1}(I_{nt}),\Gamma)\right]\end{aligned} \tag{15}$$

where $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot|\mathcal{I}_t]$. The continuation depends on investment through cash on hand y_{t+1} and through the information set $\mathcal{I}_{t+1}$. Let $\lambda_{t+1} \equiv \frac{\partial}{\partial y_{t+1}}\mathbb{E}_\theta\left[V^*(y_{t+1},\Gamma,\theta) - \bar{R}(y_{t+1},\mathcal{I}_{t+1},\Gamma)\middle|\mathcal{I}_{t+1}\right]$ denote the marginal value of cash on hand, which equals $u'(c_{t+1})$. Differentiating through y_{t+1} gives the terms in λ_{t+1}, and differentiating through the information set gives the term in $\bar{R}$. From the FOCs for consumption and assets, we get the consumption Euler equation $u'(c_t) = \mathbb{E}_t\,u'(c_{t+1}) + \kappa_{at}$. When capital constraints do not bind, $u'(c_t) = \mathbb{E}_t\,u'(c_{t+1})$. When they bind $\kappa_{at} > 0 \Rightarrow u'(c_t) > \mathbb{E}_t\,u'(c_{t+1})$. Solving for optimal investment in the safe good,

$$\begin{aligned}\delta\,\mathbb{E}_t\left[\lambda_{t+1}\frac{\partial}{\partial I_{st}}\pi_s(I_{st},\nu_{st+1})\right] &= \lambda_t w_s - \iota_{st} \\ \delta\,\mathbb{E}_t\left[\frac{u'(c_{t+1})}{u'(c_t)}\frac{\partial}{\partial I_{st}}\pi_s(I_{st},\nu_{st+1})\right] &= w_s - \frac{1}{u'(c_t)}\iota_{st} \\ \delta\left\{\mathbb{E}_t\left[\frac{u'(c_{t+1})}{u'(c_t)}\right]\mathbb{E}_t\left[\frac{\partial}{\partial I_{st}}\pi_s(I_{st},\nu_{st+1})\right] + \frac{1}{u'(c_t)}Cov_t\left(u'(c_{t+1}),\frac{\partial}{\partial I_{st}}\pi_s(I_{st},\nu_{st+1})\right)\right\} &= w_s - \frac{1}{u'(c_t)}\iota_{st}\end{aligned}$$

ι_{st} is a Lagrangian multiplier ensuring non-negative investment. Investment in the new good is

$$\begin{aligned}\delta\left\{\mathbb{E}_t\left[\frac{u'(c_{t+1})}{u'(c_t)}\right]\mathbb{E}_t\left[\frac{\partial}{\partial I_{nt}}\pi_n(I_{nt},\nu_{nt+1}+\theta)\right] + \frac{1}{u'(c_t)}Cov_t\left(u'(c_{t+1}),\frac{\partial}{\partial I_{nt}}\pi_n(I_{nt},\nu_{nt+1}+\theta)\right) - \right. \\ \left.\frac{1}{u'(c_t)}\frac{\partial}{\partial I_{nt}}\mathbb{E}_t\left[\bar{R}(y_{t+1},\mathcal{I}_{t+1}(I_{nt}),\Gamma)\right]\right\} = w_n - \frac{1}{u'(c_t)}\kappa_{\chi t}(2I_{nt}-\chi)\end{aligned}$$

given that this value satisfies the participation constraint that the utility exceeds that from investing 0, where $\iota_{nt} = 0$ whenever the new product is stocked. This check is needed since minimum order sizes create a non-convex choice set, so the entry predictions are established in Appendix A.3 by comparing values directly.

A.2 Derivation of the derivative of Bayesian regret w.r.t investment

This section shows that

$$\frac{\partial}{\partial I_{nt}} \mathbb{E}\left[\bar{R}(y_{t+1}, \mathcal{I}_{t+1}(I_{nt}), \Gamma)|\mathcal{I}_t\right] = -\frac{1}{2} \sum_{\tau=t+1}^{\infty} \delta^{\tau-t-1} Cov\left(R_\tau(\theta), \eta'_{nt+1}\Sigma_x^{-1}\eta_{nt+1}\middle|\mathcal{I}_t\right) \leq 0$$

where $\eta_{nt+1} \sim \mathcal{N}(0, I_{nt}^{-1}\Sigma_x)$ is the component of the signal $x(I_{nt}) = \theta_0 + \nu_{nt+1} + \eta_{nt+1}$ whose precision marginal investment increases. The inequality is typically strict whenever the signal changes the agent's decisions with positive probability.

Consider an agent in time t making investment I_{nt}. In period $t+1$, they learn from this investment, which will affect their Bayesian regret associated with decisions beginning in period $t+1$, $\bar{R}(y_{t+1}, \mathcal{I}_{t+1}(I_{nt}), \Gamma)$. The aim is to find

$$\frac{\partial}{\partial I_{nt}} \mathbb{E}\left[\bar{R}(y_{t+1}, \mathcal{I}_{t+1}(I_{nt}), \Gamma)|\mathcal{I}_t\right]$$

As in Appendix A.1, the derivative is taken with respect to the information set, holding y_{t+1} fixed. Realized profits depend on ν_{nt+1} but not on η_{nt+1}, so investment affects the information set only through the distribution of η_{nt+1}.

Let $c^*_\tau(\theta)$ denote the expected utility maximizing consumption path given θ and $\bar{c}_\tau(\theta)$ be consumption along the agent's planned expected utility maximizing path if θ is the truth. The planned path depends on the signals the agent receives, including $x(I_{nt})$, while $c^*_\tau(\theta)$ does not. Applying the definition of Bayesian regret and the Law of Iterated Expectations

$$\mathbb{E}\left[\bar{R}(y_{t+1}, \mathcal{I}_{t+1}(I_{nt}), \Gamma)|\mathcal{I}_t\right] = \sum_{\tau=t+1}^{\infty} \delta^{\tau-t-1}\, \mathbb{E}\left[u(c^*_\tau(\theta))\middle|\mathcal{I}_t\right] - \sum_{\tau=t+1}^{\infty} \delta^{\tau-t-1}\, \mathbb{E}\left[u(\bar{c}_\tau(\theta))\middle|\mathcal{I}_t\right]$$

The expectations are over θ, demand shocks ν_n and ν_s, the draw η_{nt+1}, and noise in future signals. This captures the fact that at t, the agent does not know what draw $x(I_{nt})$ they will receive, so they take an expectation over the possible signals, as well as expected future learning about demand (e.g.

from planned investment). The first term does not depend on η_{nt+1}, since the optimal consumption path given θ does not depend on the signal received, which has no direct effect on payoffs.

Turning to the second term, let ω denote primitive random draws other than η_{nt+1}: θ, demand shocks, the draw determining whether the agent loses access, and independent standardized draws generating the noise in future signals. The distribution of ω does not depend on I_{nt}. Future signals are constructed from these draws and the inventories prescribed by the agent's contingent plan, so holding the plan fixed, I_{nt} affects this term only through the density of η_{nt+1}, $f(\eta_{nt+1}; I_{nt})$. Then

$$\sum_{\tau=t+1}^{\infty} \delta^{\tau-t-1} \mathbb{E}\left[u(\bar{c}_\tau(\theta))\middle|\mathcal{I}_t\right] = \int \int \sum_{\tau=t+1}^{\infty} \delta^{\tau-t-1} u(\bar{c}_\tau(\theta)) f(\eta_{nt+1}; I_{nt})\, d\eta_{nt+1}\, dP(\omega)$$

Investment changes this expression in two ways: it changes the distribution of signal draws, and it changes the planned path, which responds to the signal. Holding y_{t+1} fixed, the set of feasible contingent plans does not depend on I_{nt}, and the planned path is optimal among them. By the Envelope Theorem, applied to total discounted utility rather than to each period separately, the second effect is therefore zero wherever the optimized value is differentiable in I_{nt}, which may fail at values of I_{nt} where the optimal plan changes discretely. Therefore

$$\frac{\partial}{\partial I_{nt}} \mathbb{E}\left[\bar{R}(y_{t+1}, \mathcal{I}_{t+1}(I_{nt}), \Gamma)|\mathcal{I}_t\right] = -\mathbb{E}\left[\sum_{\tau=t+1}^{\infty} \delta^{\tau-t-1} u(\bar{c}_\tau(\theta)) \frac{\partial \log f(\eta_{nt+1}; I_{nt})}{\partial I_{nt}}\middle|\mathcal{I}_t\right]$$

Differentiating the normal density with respect to I_{nt}, with $k = \dim(\theta)$ (one given the assumptions on θ, but I derive the results in matrix form for generality),

$$\frac{\partial \log f(\eta_{nt+1}; I_{nt})}{\partial I_{nt}} = \frac{k}{2I_{nt}} - \frac{1}{2}\eta'_{nt+1}\Sigma_x^{-1}\eta_{nt+1} = -\frac{1}{2}\left(\eta'_{nt+1}\Sigma_x^{-1}\eta_{nt+1} - \mathbb{E}\left[\eta'_{nt+1}\Sigma_x^{-1}\eta_{nt+1}\right]\right)$$

where the second equality leverages results about the expectation of a quadratic form, $\mathbb{E}[\eta'_{nt+1}\Sigma_x^{-1}\eta_{nt+1}] = Tr(\Sigma_x^{-1}I_{nt}^{-1}\Sigma_x) = k/I_{nt}$. Since this has mean zero, the expectation of its product with utility is a covariance. Plugging this in

$$\frac{\partial}{\partial I_{nt}} \mathbb{E}\left[\bar{R}(y_{t+1}, \mathcal{I}_{t+1}(I_{nt}), \Gamma)|\mathcal{I}_t\right] = \frac{1}{2}\sum_{\tau=t+1}^{\infty} \delta^{\tau-t-1} Cov\left(u(\bar{c}_\tau(\theta)), \eta'_{nt+1}\Sigma_x^{-1}\eta_{nt+1}\middle|\mathcal{I}_t\right)$$

Since $c^*_\tau(\theta)$ does not depend on η_{nt+1}, $Cov\left(u(c^*_\tau(\theta)), \eta'_{nt+1}\Sigma_x^{-1}\eta_{nt+1}\middle|\mathcal{I}_t\right) = 0$, and subtracting it

gives

$$\frac{\partial}{\partial I_{nt}} \mathbb{E}\left[\bar{R}(y_{t+1}, \mathcal{I}_{t+1}(I_{nt}), \Gamma)|\mathcal{I}_t\right] = -\frac{1}{2} \sum_{\tau=t+1}^{\infty} \delta^{\tau-t-1} Cov\left(R_\tau(\theta), \eta'_{nt+1}\Sigma_x^{-1}\eta_{nt+1}\middle|\mathcal{I}_t\right)$$

So the marginal reduction in regret from increasing I_{nt} is determined by the covariance between regret and the squared error of the signal. Because η_{nt+1} has variance $I_{nt}^{-1}\Sigma_x$, the expected squared error, k/I_{nt}, falls with investment. The covariance is taken over the joint distribution of η_{nt+1} and ω, so it accounts for how the agent's future stocking, and therefore the precision of its future signals, responds to $x(I_{nt})$. Demand fluctuations ν_{nt+1} do not enter the quadratic form because investment does not change their distribution. They nonetheless reduce the value of learning through the covariance: a noisier signal receives less weight in the agent's posterior, so a given error η_{nt+1} has a smaller effect on decisions and regret.

This expression is weakly negative for any increasing $u(\cdot)$: greater investment yields a more precise signal, and expected regret is weakly falling in the informativeness of $\mathcal{I}_t$ since an agent could always choose to disregard information. In particular, adding independent noise $\mathcal{N}(0, (I_{nt}^{-1} - I_{nt}'^{-1})\Sigma_x)$ to the signal from $I'_{nt} > I_{nt}$ reproduces the signal from I_{nt}, so by Blackwell's theorem the former is more informative. The covariance determines its magnitude. Intuitively, regret is typically increasing in signal noise, since a larger error moves the agent's beliefs further from the truth, making it more likely to stock when the product is unprofitable or not stock when it is profitable.

A.3 Proof of propositions

Proof of Proposition 1

I focus on the discrete choice of whether to stock at least χ units or not. This matches the mean-preserving contraction implemented in the experiment.

I show that a risk neutral firm does not respond to the mean preserving contraction. A risk averse firm does respond when beliefs fall in a range where the contraction is pivotal. Typically the response is to increase stocking, though theoretically the reverse could hold in a case where the firm perceives a sufficiently high chance that demand will be high enough that they wish to greatly increase stocking to the point capital constraints will bind, such that having slightly more cash could be valuable. This does not affect the test of risk aversion, as a positive response still only holds under risk aversion. Moreover, the empirical application makes opting into insurance

optional, so the model predicts that the implemented design can only increase stocking.

The contraction changes realized profits in period $t+1$, but lasts only one period and does not affect the signal, so stocking is equally informative under either offer. The costs of stocking I_{nt} are also unaffected. The problem therefore reduces to examining how the mean-preserving contraction affects the present value of expected utility associated with the contracted profits.

After profits are realized the firm re-optimizes. Because the feasible set $\{0\} \cup [\chi, \infty)$ is non-convex, I fix the firm's future stocking plan. Let d denote a contingent plan specifying how much of the new product the firm stocks in each future period, possibly zero, as a function of the demand realizations and signals it observes. Consumption, savings, and safe good investment are chosen optimally given d. Let $\nu_{nt+1}+\theta$ denote the demand realization from stocking χ, which is unaffected since the contraction only affects the mapping to profits, and y cash on hand in $t+1$. Let $W_d(y, \nu_{nt+1}+\theta)$ denote the present value of expected utility when the firm follows d, so that the value of stocking is $\max_d W_d(y, \nu_{nt+1}+\theta)$. Given d, the remaining choices lie in a convex set and cash enters the budget constraint linearly, so each W_d inherits the curvature of $u(\cdot)$ in y for each demand realization.

The contraction is equivalent to a transfer $\Delta(\nu_{nt+1}+\theta) = \pi_n^p - \pi_n$ that depends only on $\nu_{nt+1}+\theta$, with $\mathbb{E}_t[\Delta(\nu_{nt+1}+\theta)] = 0$. Under the original offer, cash on hand is y, and under the contraction it is $y + \Delta(\nu_{nt+1}+\theta)$.

Case 1: If the agent is risk neutral, $u'(c_t) = \bar{u}$. Then for any policy $W_d(y, \nu_{nt+1}+\theta) = \bar{u}y + G_d(\nu_{nt+1}+\theta)$, where G_d captures the continuation value of learning under the policy, which is unaffected by the contraction. The optimal policy therefore does not depend on cash, and the contraction changes expected utility by $\bar{u}\ \mathbb{E}_t[\Delta(\nu_{nt+1}+\theta)] = 0$, so the firm's utility from stocking $I_{nt} > 0$ is unaffected.

This argument requires that a firm could feasibly reduce consumption to cope with losses from stocking the product (though doing so may not be optimal for a risk averse firm), rather than cutting back on investing another product which could generate precautionary motives under a concave production function and profit maximization. This is not equivalent to assuming that capital markets are complete since agents cannot borrow freely to smooth consumption. For sufficiently large risk, this assumption would not be plausible, but in this setting shops seem empirically able to absorb losses from selling helmets through consuming fewer profits.

Case 2: If the agent is risk averse, then by definition $u''(c_t) < 0$, and each W_d is therefore strictly

concave in y, since lower cash on hand in a period lowers consumption.

Let d^* be the optimal plan absent the contraction. Given the mean-preserving contraction, d^* is still feasible (typically π^p would pay out more when realized profits are low, and so feasibility is more easily satisfied under this contract, though by the argument above this is not needed since a firm could always cut back on consumption to stock). Because the firm can re-optimize, the contraction raises expected utility by at least

$$\mathbb{E}_t \left[W_{d^*} \left(y + \Delta, \nu_{nt+1} + \theta \right) - W_{d^*} \left(y, \nu_{nt+1} + \theta \right) \right]$$

where $\Delta = \Delta(\nu_{nt+1} + \theta)$ and y denotes cash on hand under the original offer. By strict concavity of W_{d^*} in cash, its tangent at post-transfer cash lies above it, so for each demand realization

$$W_{d^*} \left(y + \Delta, \nu_{nt+1} + \theta \right) - W_{d^*} \left(y, \nu_{nt+1} + \theta \right) \geq \frac{\partial W_{d^*}}{\partial y} \left(y + \Delta, \nu_{nt+1} + \theta \right) \Delta.$$

This inequality is strict wherever $\Delta \neq 0$, and so in expectation, since the contraction differs from the original offer. Since $\mathbb{E}_t[\Delta] = 0$, the expectation of the right hand side equals its covariance, so the gain in utility from the contraction strictly exceeds

$$Cov_t \left(\frac{\partial W_{d^*}}{\partial y} \left(y + \Delta, \nu_{nt+1} + \theta \right), \Delta \right).$$

Because the contraction lowers the variance of profits while preserving their mean, Δ is negatively correlated with realized profits: the contraction pays out more, on average, when profits are low.[23] By concavity, cash is also more valuable when profits are low. This typically leads the contraction to move cash toward states in which it is more valuable, making the covariance above positive and indicating that the contraction raises the agent's utility from stocking. This can fail if the agent expected, with high enough probability, news favorable enough to make it wish to expand substantially while facing binding liquidity constraints, so that cash is more valuable after high demand despite the lower losses the contraction provides when demand is low. Any such effect does not affect Case 1, so it cannot generate a spurious rejection of risk neutrality.

Therefore a mean-preserving contraction cannot lead a risk neutral firm to stock, while it can lead a risk averse firm with beliefs at which it is pivotal to stock when it otherwise would not. Since $u(\cdot)$ is weakly concave, an agent that is not risk neutral is risk averse, so the contraction leads a firm

[23] Writing $\pi_n^p = \pi_n + \Delta$, $Var(\pi_n^p) < Var(\pi_n)$ implies $Cov(\pi_n, \Delta) < -Var(\Delta)/2 < 0$.

to stock only if it is risk averse.

Finally, let $\pi_n^{p'}$ be any offer that pays out weakly less than π_n^p in every state, which includes the offer used in the study. Any agent that prefers $\pi_n^{p'}$ to π_n also prefers π_n^p to π_n, and so must be risk averse by the argument above.

Proof of Proposition 2

Part a: Realized profits are weakly higher in every state under returns, while the signal generated by stocking and the cost of stocking are unchanged. The present value of expected utility from stocking is increasing in realized profits for any increasing utility function, so the result follows immediately.

Part b: This result follows from the fact that the continuation value of learning reflects the fact that a firm which experiments can restock if demand is high and exit the market if it is low. Resolving future restocking uncertainty therefore provides stronger incentives for a forward looking agent to experiment to discover if the product is profitable.

More formally, and in terms of participation, any strategy with stocking is worth weakly more when access is guaranteed, as a firm that retains access could always choose to exit after t_c. Furthermore, an agent that stocks today expects to enter period t_c with less uncertain beliefs than one that delays, and so by Blackwell's information criterion the option value of continuing to stock yields weakly higher utility, and strictly so if the agent's optimal path involves stocking after that point.

If access is guaranteed and the agent is not capital constrained, then delaying experimentation yields neither information nor a financial gain, whereas experimenting sooner gives the firm more time to act on information.[24] Hence, under guaranteed access any strategy involving stocking starting in period $t' > t$ would yield higher utility if started today. Since the value of never stocking does not depend on Γ, any firm that stocks under uncertain access also stocks when access is guaranteed, and for some set of beliefs the firm will stock under the resolution of Γ but not otherwise.

Under binding capital constraints, a firm may plan to wait until constraints are slack to stock. An increase in Γ may incentivize experimenting earlier in such cases, but will not induce firms to stock that otherwise would never do so. Furthermore, a fall in Γ does not change the value of never stocking, since it only affects stocking access. Therefore a reduction in Γ weakly raises all firms'

[24] Technically under risk aversion, a firm could wish to wait to gain wealth. This is not plausibly consequential over the time horizon studied, so I abstract from this possibility.

propensity to experiment at some date, and will strictly increase the share of shops that stock in a population including firms that are unwilling to stock with high Γ but willing to with low Γ.

If the firm faces no uncertainty about demand, $\Sigma_t = 0$, then there is no continuation value of learning so a change in Γ does not affect stocking decisions.

Part c: Capital constraints not binding gives $\mathbb{E}_t\left[\frac{u'(c_{t+1})}{u'(c_t)}\right] = 1$, and with $\zeta_n(\chi) = 0$ returns guarantee $p_{nt+1} \geq w_n$ so that $\pi_n(\chi, \nu_{nt+1} + \theta) - w_n\chi = \chi \cdot p_n(\chi, \nu_{nt+1} + \theta) - \chi \cdot w_n \geq 0$ in every state.

Suppose $Pr(\pi_n(\chi, \nu_{nt+1} + \theta) > w_n\chi) > 0$. With returns, stocking χ costs the firm at most the time value of the refund, $(1-\delta)w_n\chi$, while yielding the upside $\mathbb{E}_t[(\pi_n(\chi, \nu_{nt+1} + \theta) - w_n\chi)^+] > 0$ and a signal about θ. For given beliefs, stocking therefore dominates not stocking for δ sufficiently close to 1, regardless of risk preferences, so a change in Γ cannot alter whether an agent stocks at the extensive margin.

If instead $Pr(\pi_n(\chi, \nu_{nt+1} + \theta) > w_n\chi) = 0$, then regret is 0 and there is no learning value, so changes in Γ do not affect decisions and the agent never stocks. Therefore, if offered returns that eliminate loss risk, $\frac{\partial}{\partial\Gamma}\mathbb{I}(I^*_{nt} > 0) = 0$.

Part d: A one-time return offer causes the firm to stock and observe a signal, which shrinks Σ_{t+1} and so alters subsequent stocking even after the offer is withdrawn. The sign of the effect depends on risk preferences and on how beliefs update. If agents are risk neutral and have correctly centered beliefs, then expected profits from stocking χ are unaffected but learning value is lower so investment falls. If expected profits positively update or agents are sufficiently risk averse, then the expected utility of stocking increases so there will be persistent positive effects on stocking.

Proof of Proposition 3

Parts a.i and b.i: Consider a risk neutral agent. An increase in prior uncertainty, holding mean beliefs fixed, increases the upside from experimentation since an agent may discover that a product is highly profitable and then can restock, whereas they may exit the market if losses are high. In particular, an increase in prior uncertainty will generally raise regret from never stocking, making stocking more attractive.

Formalizing this argument in terms of the participation condition, a firm experiments if there exists a plan with stocking that beats never stocking. Beliefs about demand may be written as $\theta = \mu_t + \sqrt{\Sigma_t}Z$, where $Z \sim \mathcal{N}(0,1)$.

Consider two priors with the same μ_t but variances $\Sigma_L < \Sigma_H$. A signal with variance V reveals Z with noise variance V/Σ_t, so the less uncertain firm's signals are a garbling of the more uncertain firm's, reflecting that a more uncertain firm's posteriors are moved more by a signal with fixed variance. Blackwell's information criterion implies the more uncertain firm can follow any plan available to the less uncertain firm.

Because profits are affine in θ, expected discounted profits from a plan d can be written $J_d(\Sigma) = A_d + \sqrt{\Sigma}B_d$, where A_d collects expected costs and revenues at mean demand, and B_d captures possible profits from restocking if demand turns out to be high. For the optimal plan, $B_d \geq 0$. Intuitively, the firm restocks after good not bad news. Formally, let d' be the plan that does after each signal what d does after the opposite signal, stocking after bad news wherever d stocks after good news. Because good and bad news are equally likely, d' stocks as often as d and has the same expected costs, but earns $A_d - \sqrt{\Sigma}B_d$. Since d is optimal, it earns at least as much as d', which requires $B_d \geq 0$. Following the less uncertain firm's optimal plan therefore earns the more uncertain firm $(\sqrt{\Sigma_H} - \sqrt{\Sigma_L})B_d \geq 0$ more, and it can do at least as well with its own plan, so the value of experimenting is increasing in Σ_t. Since the value of never stocking does not depend on Σ_t, and a firm that does not stock learns nothing before it begins stocking, any firm that experiments under Σ_L also experiments under Σ_H, giving a.i.

For part b.i, consider a risk averse agent. Because profits are affine in θ, a more diffuse prior with the same mean spreads out the profits from stocking χ without changing their expected value. By the argument in the proof of Proposition 1, this lowers a risk averse agent's expected utility from stocking, and this can outweigh any increase in the value of experimentation. For a sufficiently risk averse agent, greater uncertainty therefore lowers the value of stocking relative to not stocking, reversing part a.i.

Parts a.ii and b.ii: Returns do not affect the informativeness of the signal generated by stocking χ, and regret is a function of payoffs beginning in period $t+2$, which depend on period $t+1$ payoffs only through cash on hand y_{t+1}.

For a risk neutral agent, the value of stocking absent returns is increasing in μ_t, since expected profits rise with expected demand, and in Σ_t, by part a.i. A firm that stocks without returns therefore has beliefs at which this value is non-negative, while a firm that stocks only with returns has beliefs at which it is negative. Holding other primitives fixed, a firm induced to stock by returns cannot have

both a weakly higher prior mean and a weakly higher prior variance than a firm that stocks without returns, so it must be more pessimistic or more certain about demand, giving part a.ii. In particular, among firms with the same expected demand, those induced to stock by returns are less uncertain than those that stock without them. Note that this argument does not use the location of priors relative to θ_0, so it applies to risk neutral learners with correctly centered as well as pessimistic beliefs.

For a risk averse agent, returns raise payoffs in the states where stocking would otherwise make a loss, which is more valuable the more likely and larger those losses are. Because profits are affine in θ, a more diffuse prior with the same mean spreads out profits, making losses more likely and larger. The expected utility gain from returns is therefore larger for firms with more diffuse priors, giving part b.ii.

Parts a.iii and b.iii: Stocking in period t lowers the agent's uncertainty, so $\Sigma_{t+1} < \Sigma_t$. For a risk neutral agent, the gain from stocking again in period $t+1$ is expected profits plus the value of experimentation. Since greater uncertainty raises the value of experimentation (part a.i), lower uncertainty reduces this value, so a risk neutral agent obtains a lower gain from stocking again unless its expected profits rise. Persistent stocking by a risk neutral agent therefore requires that expected profits positively update.

If agents are sufficiently risk averse, condition on the realization of the posterior mean μ_{t+1}. Holding μ_{t+1} fixed, the reduction from Σ_t to Σ_{t+1} is a mean-preserving contraction of the distribution of $\pi_n(\chi, \nu_{nt+2}+\theta)$ and so raises expected utility from stocking. An agent may therefore obtain higher expected utility from stocking $I_{nt+1} = \chi$ even if expected profits are unchanged or fall.

Proof of Proposition 4

Part a: From Equation 1, past learning from peers lowers Σ_t. Returns raise the payoff from stocking in states where it would otherwise make a loss. Because profits are affine in θ, a lower Σ_t (holding mean beliefs fixed) makes these losses smaller in expectation, so returns are less valuable. Supply chain uncertainty affects the gain from stocking only through learning value. As Σ_t falls to 0, a change in Γ no longer affects the gain from stocking.

The result that past learning from peers should mimic the effect of learning from one's own stocking follows from the fact that both signals are centered on the truth and reduce uncertainty, varying only precision.

Part b: Exposure to a seller generates a signal x_{ot} with precision Σ_o^{-1}, which lowers Σ_t and moves μ_t toward the signal. The lower uncertainty reduces the risk of stocking, which raises a risk averse agent's expected utility from stocking by the argument in Proposition 1. If the agent's prior is pessimistic, signals are centered on the true θ_0, which lies above its prior mean, so exposure tends to raise the posterior mean and the value of stocking. In both cases the lower uncertainty also reduces the value of experimentation, so exposure raises stocking when the agent is sufficiently risk averse or its prior is sufficiently pessimistic.

Part c: Suppose the agent receives a signal x_{ot} below its expectation, so that μ_t falls while Σ_t also falls. For a risk neutral agent, a lower μ_t lowers the value of stocking, since any stocking strategy earns less when expected demand is lower. A lower Σ_t also lowers the value of stocking when greater uncertainty raises the value of experimentation, as in Proposition 3. A risk neutral agent therefore stocks less after such a signal. The fall in Σ_t also reduces the risk of stocking, which raises a risk averse agent's expected utility from stocking but has no effect when $u'(\cdot)$ is constant. Receipt of such a signal can therefore raise stocking only if the agent is risk averse.

A.4 Proof that the insurance contract induces a mean-preserving contraction

First suppose that the premium payments to enterprises are

$$P_i = 1000 \cdot (1 - so_i)$$

and let the insurance contract be as given, paying out $1,000$ if sales are less than 3 and 0 if the shop sells 3 helmets. The premium P_i is strictly lower for all so_i than the premium used in the study, so the control offer in the study dominates the offer studied here. Proving that the treatment offer is a mean-preserving contraction under the P_i here is thus sufficient to conclude that the insurance offer increases investment only if firms are risk averse.

The restriction that shops cannot restock if they accept the insurance payout ensures that intentionally selling fewer helmets is not profit increasing. Withholding a sale is only problematic when demand is strong enough to sell out, in which case selling out and restocking would yield higher profits versus collecting the payout and waiting to sell the final helmet. Moreover, prices were no higher on average among those that opted into insurance, those with insurance sold out more often than anticipated, and all enterprises passed audits.

Let $\pi_n(3, \nu_{nt+1} + \theta)$ be profits, inclusive of P_i and net of stock costs, under the control offer and $\pi_n^p(3, \nu_{nt+1} + \theta)$ be these profits under the treatment offer. By construction, and recalling that $so_i = Pr(\text{sales}= 3)$,

$$\begin{aligned}
\mathbb{E}_t[\pi_n^p(3, \nu_{nt+1} + \theta)] &= Pr(\text{sales} < 3) \cdot (1000 - P_i + \mathbb{E}_t[\pi_n(3, \nu_{nt+1} + \theta)|\text{sales} < 3]) \\
&\quad + Pr(\text{sales} = 3) \cdot (-P_i + \mathbb{E}_t[\pi_n(3, \nu_{nt+1} + \theta)|\text{sales} = 3]) \\
&= \underbrace{1000 \cdot (1 - so_i)}_{=P_i} - P_i + Pr(\text{sales} < 3) \cdot \mathbb{E}_t[\pi_n(3, \nu_{nt+1} + \theta)|\text{sales} < 3] \\
&\quad + Pr(\text{sales} = 3) \cdot \mathbb{E}_t[\pi_n(3, \nu_{nt+1} + \theta)|\text{sales} = 3] \\
&= \mathbb{E}_t[\pi_n(3, \nu_{nt+1} + \theta)]
\end{aligned}$$

where the last line leverages the Law of Total Probability. Therefore the expected profits are the same under the two offers. Assume that $0 < so_i < 1$ since trivially the offers are the same if the agent perceives no risk of failing to sell out.

Let h_i be the price charged for a helmet, and assume

$$h_i > \max\{P_i,\ 1000 - P_i\} = 1000 \cdot \max\{so_i,\ 1 - so_i\}.$$

The second term always holds empirically. The first condition holds in most cases, only failing in a minority of cases where firms have very pessimistic beliefs (and only one goes on to stock). In these cases, the premium payment approaches the insurance payout, so insurance becomes unattractive under any set of risk preferences (e.g. if $P_i \approx 1,000$).

Let F denote the CDF of profits under π_n and F_p under π_n^p, and let $\mathbb{P}_s = Pr(\text{sales} = s)$ for $s \in \{0, 1, 2\}$, so that $\mathbb{P}_0 + \mathbb{P}_1 + \mathbb{P}_2 = 1 - so_i$. Payouts change only when demand crosses a threshold, so F jumps at P_i, $h_i + P_i$, $2h_i + P_i$ and $3h_i + P_i$, and F_p jumps at 1000, $h_i + 1000$, $2h_i + 1000$ and $3h_i$. Note that the last of these need not be the largest: when $h_i < 1000$ the sell-out payoff $3h_i$ falls below $2h_i + 1000$, so the jumps of F_p occur in the order 1000, $h_i + 1000$, $3h_i$, $2h_i + 1000$.

For $x < 3h_i$ we have $F_p(x) \leq F(x)$, since $P_i < 1000$ places every jump of F weakly to the left of the corresponding jump of F_p and $3h_i$ exceeds $2h_i + P_i$ by assumption. Hence $\int_{-\infty}^{x} F_p(y)dy \leq \int_{-\infty}^{x} F(y)dy$ on this range, with strict inequality at $x = 2h_i + P_i$ whenever $\mathbb{P}_0 + \mathbb{P}_1 > 0$. For $y \geq 3h_i + P_i$ we have $F_p(y) = F(y) = 1$. It therefore suffices to establish the inequality on $[3h_i, 3h_i + P_i]$.

On this interval $F(y) = 1 - so_i$ throughout, because $P_i + 2h_i < 3h_i$ and $P_i + 3h_i$ is the right

endpoint. Meanwhile F_p is weakly increasing, so $\int_{-\infty}^{x}(F_p - F)$ is piecewise linear in x with non-decreasing slope, and hence convex, on $[3h_i, 3h_i + P_i]$. It is weakly negative at $x = 3h_i$ by the argument above, and equals zero at $x = 3h_i + P_i$ because the two offers have the same mean. A convex function that is weakly negative at the left endpoint of an interval and zero at the right endpoint is weakly negative throughout. It therefore suffices to verify that $\int_{-\infty}^{3h_i+P_i} F_p(y)dy \leq \int_{-\infty}^{3h_i+P_i} F(y)dy$. First,

$$\begin{aligned}\int_{-\infty}^{3h_i+P_i} F(y)dy &= \int_{-\infty}^{P_i} F(y)dy + \int_{P_i}^{P_i+h_i} F(y)dy + \int_{P_i+h_i}^{P_i+2h_i} F(y)dy + \int_{P_i+2h_i}^{P_i+3h_i} F(y)dy \\ &= h_i \cdot \mathbb{P}_0 + h_i \cdot (\mathbb{P}_0 + \mathbb{P}_1) + h_i \cdot (\mathbb{P}_0 + \mathbb{P}_1 + \mathbb{P}_2) \\ &= h_i \cdot (3\mathbb{P}_0 + 2\mathbb{P}_1 + \mathbb{P}_2).\end{aligned}$$

Case 1: Suppose that $h_i \geq 1,000$, so that $3h_i$ is the largest jump of F_p.

$$\begin{aligned}\int_{-\infty}^{3h_i+P_i} F_p(y)dy &= \int_{-\infty}^{1000} F_p(y)dy + \int_{1000}^{1000+h_i} F_p(y)dy + \int_{1000+h_i}^{1000+2h_i} F_p(y)dy \\ &\quad + \int_{1000+2h_i}^{3h_i} F_p(y)dy + \int_{3h_i}^{3h_i+P_i} F_p(y)dy \\ &= h_i \cdot \mathbb{P}_0 + h_i \cdot (\mathbb{P}_0 + \mathbb{P}_1) + (h_i - 1000) \cdot (\mathbb{P}_0 + \mathbb{P}_1 + \mathbb{P}_2) + \underbrace{P_i}_{=1000\cdot(\mathbb{P}_0+\mathbb{P}_1+\mathbb{P}_2)} \\ &= h_i \cdot (3\mathbb{P}_0 + 2\mathbb{P}_1 + \mathbb{P}_2).\end{aligned}$$

Case 2: Suppose that $1000 - P_i < h_i < 1000$,

$$\begin{aligned}\int_{-\infty}^{3h_i+P_i} F_p(y)dy &= \int_{-\infty}^{1000} F_p(y)dy + \int_{1000}^{1000+h_i} F_p(y)dy + \int_{1000+h_i}^{3h_i} F_p(y)dy \\ &\quad + \int_{3h_i}^{2h_i+1000} F_p(y)dy + \int_{2h_i+1000}^{3h_i+P_i} F_p(y)dy \\ &= h_i \cdot \mathbb{P}_0 + (2h_i - 1000) \cdot (\mathbb{P}_0 + \mathbb{P}_1) + (1000 - h_i) \cdot (\mathbb{P}_0 + \mathbb{P}_1 + so_i) \\ &\quad + \big(h_i + P_i - 1000\big) \\ &= h_i \cdot (3\mathbb{P}_0 + 2\mathbb{P}_1 + \mathbb{P}_2). \quad \square\end{aligned}$$

B Coefficient of relative risk aversion estimation

This section estimates the coefficient of relative risk aversion, ρ, of firms in the Insurance Experiment. I caution that the estimates rely on strong assumptions and are generally imprecise. This is driven largely by the fact that the Insurance Experiment calibrated contracts based on firms' be-

liefs about the probability of selling out, a relatively simple object that experience implementing the Learning Experiment showed shops could effectively forecast. But CRRA estimates depend on a firm's full belief distribution as measured by the frequentist approach, which is measured with substantially more noise.

I therefore view this exercise as exploratory analysis examining whether firms' behavior can be rationalized by levels of risk aversion often documented among individuals, or whether behavior seems to imply higher levels of risk aversion. This is largely exploratory and the core argument of the paper, that relaxing risk meaningfully changes firms' investment decisions, holds independently of the results presented in this appendix. I first estimate a simple structural model that aims to identify CRRA coefficients from treatment effects of insurance on stocking. I do this under (1) a narrow bracketing assumption that ignores background income (which the lottery choice game imposes), and (2) a broad bracketing case including other firm income. Second, I estimate back of the envelope bounds of *insurance* adopters risk aversion based on their decision to opt into the contract. None of the cases allow for borrowing, which is plausible since many firms report never borrowing for their enterprise in this setting. Implied risk aversion would be higher given consumption smoothing across periods, likely requiring deviations from neoclassical risk aversion to rationalize (Rabin, 2000). On the other hand, firms typically use profits to support households with more than one person, which would lead this analysis to overstate utility curvature.

A firm i decides whether to stock 3 motorcycle helmets. Stocking costs K, which a firm that does not order consumes. Conditional on ordering, the firm sells $k \in \{0, 1, 2, 3\}$ helmets, with subjective probabilities b_{ik}. These values are measured empirically from the frequentist mechanism detailed in the text. Each helmet sold yields revenue γ_i, which is measured from intended sales prices in the baseline survey.

Firms are offered premiums P_i under the *insurance* offer described in the text, which pays B if $k_i \leq 2$. Consumption is given by

$$c_i^0 = P_i + K + \omega_i, \tag{16}$$

$$c_{ik}^1 = P_i + k\,\gamma_i + \omega_i, \tag{17}$$

$$c_{ik}^2 = k\,\gamma_i + B\,\mathbf{1}\{k \leq 2\} + \omega_i, \tag{18}$$

for the no-stock, stock-without-insurance and stock-with-insurance options. ω_i gives background firm profits, set to zero in the narrow bracketing case. Flow utility is CRRA,

$$u(c;\rho) = \frac{c^{1-\rho}}{1-\rho}, \qquad \rho \neq 1, \qquad u(c;1) = \ln c. \tag{19}$$

For each stocking and insurance choice, $j \in \{0,1,2\}$, define expected utility and certainty equivalent payouts,

$$EU_i^j(\rho) = \mathbb{E}_{\omega_i}\left[\sum_{k=0}^{3} b_{ik}\, u\Big(c_{ik}^j;\rho\Big)\right], \qquad CE_i^j(\rho) = u^{-1}\Big(EU_i^j(\rho);\rho\Big), \tag{20}$$

where $u^{-1}(v;\rho) = [(1-\rho)v]^{1/(1-\rho)}$ for $\rho \neq 1$ and $\exp(v)$ for $\rho = 1$. Expectations over ω_i capture the fact that background firm income may be (and empirically is) stochastic, lowering the risk aversion needed to rationalize insurance decisions. A control firm can stock without insurance or not stock. A treated firm can opt into insurance. I assume a firm selects insurance when available optimally. The value of stocking is

$$CE_i^{\mathrm{S}}(\rho) = \begin{cases} CE_i^1(\rho), & T_i = 0, \\ \max\left\{CE_i^1(\rho),\ CE_i^2(\rho)\right\}, & T_i = 1, \end{cases} \tag{21}$$

The value of stocking is subject to unobserved idiosyncratic variation in payouts ϵ_i. The firm stocks if $\alpha + CE_i^{\mathrm{S}}(\rho) + \varepsilon_i > CE_i^0(\rho)$ with ε_i standard logistic, so

$$\Pr(y_i = 1) \;=\; \Lambda\Big(\alpha + CE_i^{\mathrm{S}}(\rho) - CE_i^0(\rho)\Big), \qquad \Lambda(z) = \frac{1}{1+e^{-z}}. \tag{22}$$

I estimate the model using certainty equivalent payouts in PPP adjusted terms since I lack variation to identify the scale of ϵ_i. Using a taste shock to expected utility with fixed scale causes idiosyncratic noise to dominate the value of stocking as risk aversion grows, creating an identification challenge for ρ. Estimating the model in certainty equivalent payouts prevents such noise from dominating, though the scale normalization still matters. This limitation, along with power limitations, motivates the back of the envelope calculations of ρ at the end of this section.

α absorbs features such as capital constraints or shelf space that may shift the average rate of stocking across firms. I fit the model to stocking within 24 hours, as the final order variable allows firms to search for customers which may distort the level of risk aversion estimated.

There are two unknowns, ρ and α. I match two moments of the data: the overall share of firms

that stock, and the difference in stocking rates between treated and control firms.

I estimate the model in two steps using GMM, using treatment assignment as an instrument. First, fix a candidate ρ. The predicted stocking rate $n^{-1}\sum_i \Lambda(\alpha + CE_i^{\mathrm{S}}(\rho) - CE_i^0(\rho))$ is strictly increasing in α, so a unique α fits the average stocking rate $\bar{y}$. Denote this by $\alpha(\rho)$. Given the candidate ρ, one may define the moment

$$\bar{g}_2(\rho) \;=\; \frac{1}{n}\sum_{i=1}^{n} T_i\left[\, y_i - \Lambda\Big(\alpha(\rho) + CE_i^{\mathrm{S}}(\rho) - CE_i^0(\rho)\Big)\right], \tag{23}$$

a treatment indicator times the difference between observed and predicted stocking. The estimate $\hat{\rho}$ is the value minimizing $|\bar{g}_2(\rho)|$ over a grid. No specification yields $\bar{g}_2(\rho) = 0$, so each estimate uses the minimum. The binary treatment indicator is a relatively weak instrument since the problem is non-linear, so I construct weak instrument robust confidence sets using Anderson-Rubin inversion. For each candidate value $\tilde{\rho}$, I refit the intercept at $\tilde{\rho}$ exactly as in estimation, compute the shortfall $\bar{g}_2(\tilde{\rho})$ from (23), and compare it to its own standard error,

$$t(\tilde{\rho}) = \frac{\bar{g}_2(\tilde{\rho})}{\widehat{\mathrm{se}}(\tilde{\rho})}, \qquad C_{95} = \left\{\, \tilde{\rho} : |t(\tilde{\rho})| \le 1.96 \,\right\}, \tag{24}$$

evaluated over $(0, 24]$. Standard errors account for $\hat{\alpha}$ having been re-estimated at each $\tilde{\rho}$.[25]

A sample of 340 firms has non-missing belief data. In the narrow bracket case, I estimate $\hat{\rho} = 0.89$ across the full sample, $\hat{\rho} = 0.64$ among those that exhibit below median risk aversion on the lottery choice game, and $\hat{\rho} = 2.85$ among those that exhibit above-median risk aversion on the game. While suggestive, these figures align well with the game and suggest that modest risk aversion may rationalize the effects under narrow bracketing.

I estimate four versions of the broad bracket case. I consider uncertainty in background income versus not, and I consider the sample that reported baseline profits only and the sample of 340 for both cases, imputing missing variables using the median.

For the case where I allow for uncertainty in background income, let $\omega_{ij} = \Pi_i \eta_j$, where Π_i is firm i's baseline monthly profit and η_j is a multiplier drawn from the empirical distribution of endline-to-baseline profit ratios, truncated at three. The ratios are taken from control firms only, so that the background distribution is not shifted by treatment. All three branches are averaged over

[25] Write $e_i = y_i - \Lambda(\alpha(\tilde{\rho}) + \Delta_i(\tilde{\rho}))$ where $\Delta_i(\rho) = CE_i^{\mathrm{S}}(\rho) - CE_i^0(\rho)$ and $\lambda_i = \Lambda(1-\Lambda)$ evaluated at $\tilde{\rho}$. $\widehat{\mathrm{se}}(\tilde{\rho})^2 = n^{-2}\sum_i \psi_i^2$ with $\psi_i = e_i\,(T_i - \bar{T}_\lambda)$ and $\bar{T}_\lambda = \sum_i T_i\lambda_i / \sum_i \lambda_i$, the treated share weighted by how responsive a firm is at $\tilde{\rho}$.

$J = 500$ draws. In the case where I shut down background income volatility, I set $\omega_i = \Pi_i \bar{\eta}$ with probability one, where $\bar{\eta}$ is the mean of the ratio distribution.

Allowing for uncertainty in background income, $\hat{\rho}$ increases to 1.76 among those without missing income data and 3.71 with missing observations imputed. Point estimates suggest that background volatility in income helps rationalize the value of insurance: treating other firm income as deterministic raises these values to 2.08 and 7.40 respectively.

The experiment was designed to test if shielding firms from risk meaningfully moved behavior, not identify ρ. As a result, weak instrument robust confidence sets typically span the full grid. Given this, I also perform a back of the envelope check in which I infer how much risk aversion would be needed for opting into insurance to be optimal among firms that selected it.

Conditional on ordering, the contract is a lottery: the firm gives up the cash payout P_i and receives B only if it does not sell out. Let $\bar{k}_i = \mathbb{E}[k \mid k \leq 2]$ be expected sales when the firm does not sell out, so revenue is $\bar{k}_i \gamma_i$ in that case and $3\gamma_i$ when it does, and let ω_i denote consumption from all other sources. Then

	not sold out (prob. $1 - so_i$)	sold out (prob. so_i)
keep the cash	$\omega_i + \bar{k}_i \gamma_i + P_i$	$\omega_i + 3\gamma_i + P_i$
take the contract	$\omega_i + \bar{k}_i \gamma_i + B$	$\omega_i + 3\gamma_i$

The contract trades a smaller unconditional premium for a larger conditional payout when helmet revenue is low. Given each firm's beliefs, there exists some ρ_i above which they would prefer insurance. For simplicity, I assume that a firm that does not sell out receives its expected sales conditional on not selling out, $\bar{k}_i$, which limits reliance on the full belief distribution. By Jensen's inequality, accounting for variation in sales when the firm does not sell out would raise the value of insurance and lower the curvature needed to rationalize taking it. The thresholds ρ_i may therefore overstate the risk aversion required.[26]

In the narrow bracket case, required curvature averages 0.42 across takers, with a median of 0.27 and a maximum of 1.15, so the structural estimate of 0.89 falls within the range these choices require. Using own baseline profits as background income, the thresholds average 4.80 with a median of 3.17. These values fall between the two broad bracket structural estimates without background volatility, suggesting consistency with those values.

[26] One insurance taker's decision cannot be rationalized at any curvature so they are dropped from this analysis because they reported that they would definitely sell out.